



Bangladesh University of Engineering & Technology

Project Report

Course No: EEE 208(S)

Course Name: Electronics Circuits II Laboratory

Project Title: “An Analog Calculator (2-digit input) using Op-Amp”

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INTRODUCTION

Project title: “An Analog Calculator using Op- Amp”

In the realm of computational technology, the choice between analog and digital computation has always been a matter of trade-offs. While digital computation offers precision and accuracy, analog computation is often celebrated for its remarkable speed. However, it is a well-known fact that analog systems tend to exhibit a certain level of inaccuracy, which has historically limited their practical applications in fields requiring precision. The delicate balance between the speed and accuracy of analog computation has been a challenge that engineers and researchers have grappled with for decades.

The objective of this project is to bridge this gap by designing and developing an analog calculator capable of taking 2-digit input and working within reasonable tolerances. Leveraging the inherent speed advantages of analog computation while ensuring accuracy to an acceptable degree is the crux of our endeavor. To achieve this goal, we have employed standard operational amplifier configurations as the foundation of our analog calculator circuits, meticulously fine-tuned to perform within specified tolerances.

This project report delves into the details of our journey towards creating an analog calculator that marries the swiftness of analog computation with the accuracy demanded by modern applications. We will discuss the principles behind operational amplifier-based circuits, the challenges we encountered, the solutions we devised, and the results we achieved and the limitations of our designed calculator. Furthermore, we will explore potential applications of our analog calculator in various domains and its significance in enhancing computational capabilities.

PROBLEM STATEMENT(OBJECTIVE):

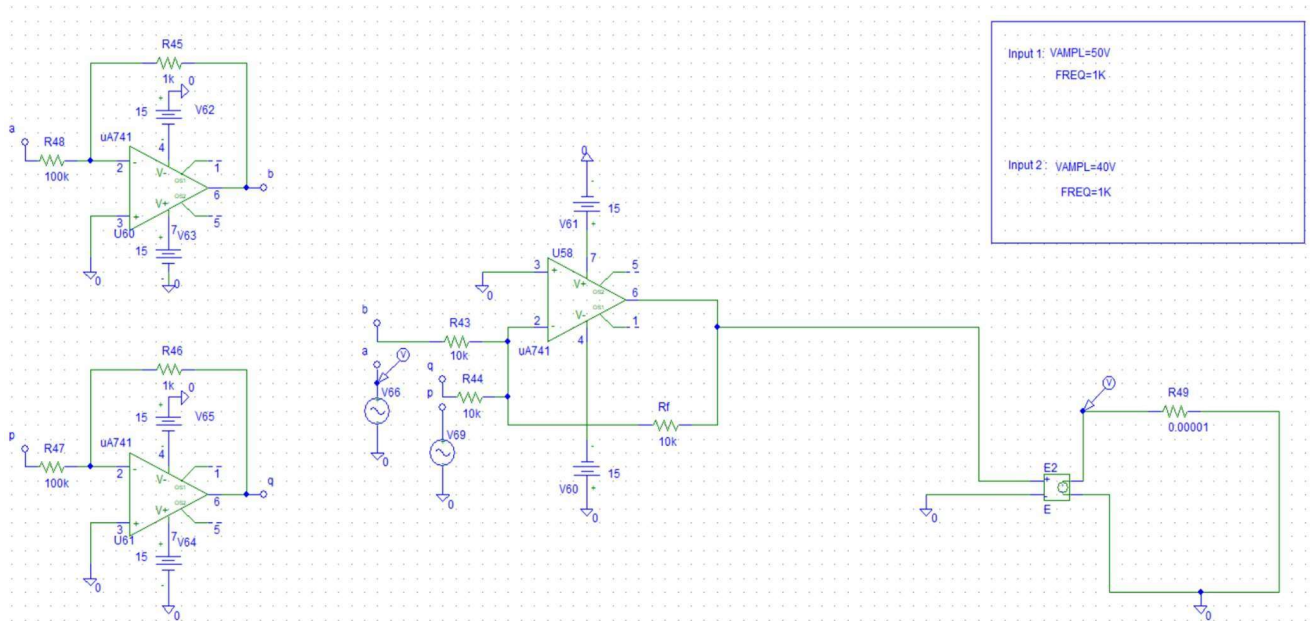
An Analog Calculator has to be built that can perform the following actions for both ac and dc ~~ip~~ signals:

1. Summation
2. Subtraction
3. Multiplier
4. Divider
5. Differentiator
6. Integrator
7. Logarithm (Base 10)
8. Natural Logarithm
9. Exponential
10. Power
11. Root

OPERATIONS (WITH CIRCUITS & OUTPUTS)

SUMMATION:

CIRCUIT AND OUTPUT(AC):



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit facilitates the addition of two input signals. Both U60 and U61 serve as inverting amplifiers, effectively dividing the input signal by 100. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

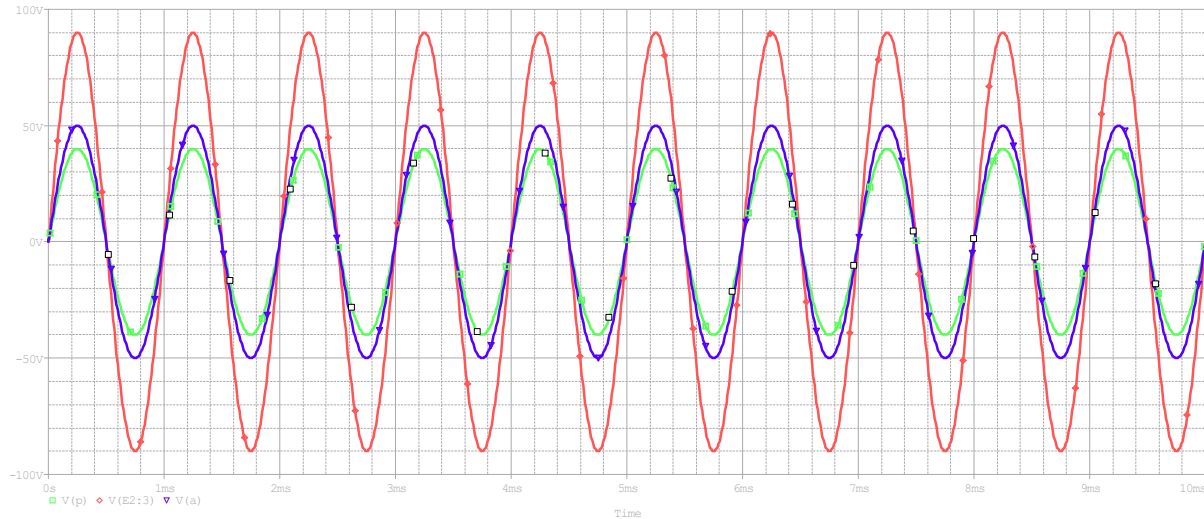
Within the circuit, operational amplifier U58 operates as an inverting summing amplifier, responsible for generating the ultimate output signal. Subsequently, this output signal undergoes a substantial 100-fold amplification through a voltage-controlled voltage source, ultimately restoring it to its original form.

Output:

Case 1:

Input 1: 50V p Sine wave, 1kHz

Input 2: 40V p Sine wave, 1kHz

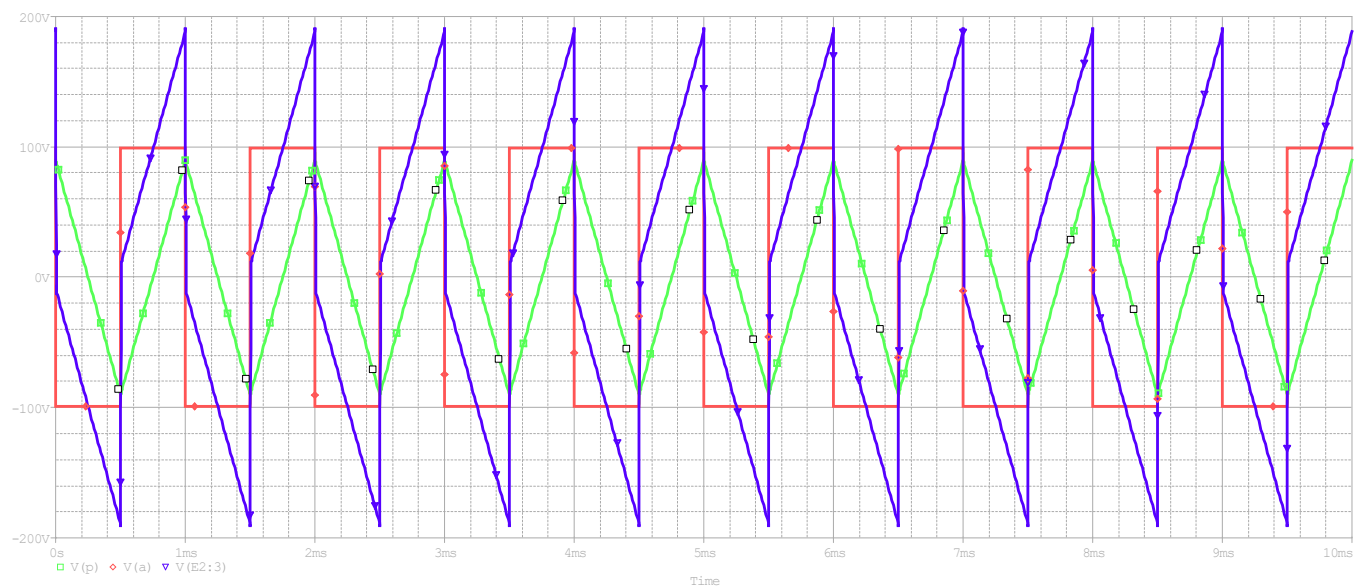


Here, blue and green are the input & red is output graph.

Case 2:

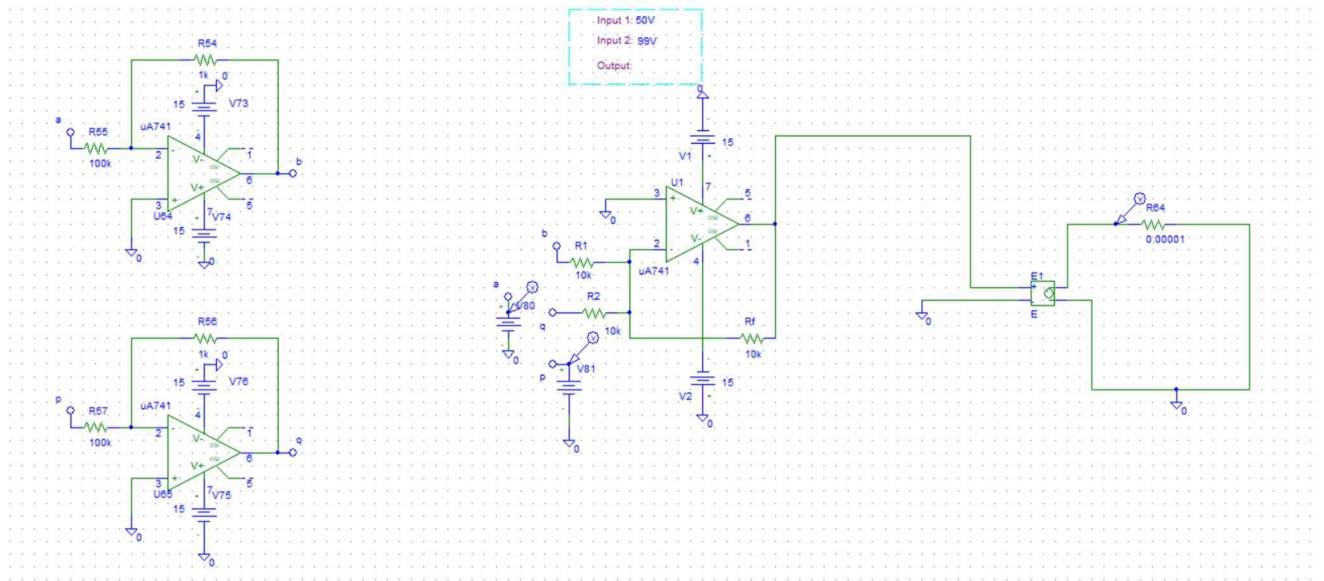
Input 1: 99V Square wave, 1kHz &

Input 2: 90V Triangular wave, 1kHz



Here, red and green are the input & blue is output graph

CIRCUIT & INPUT: (DC)

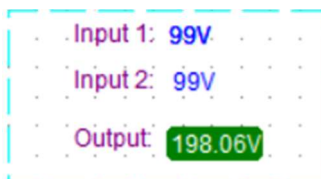


Output:

Case 1:



Case 2:

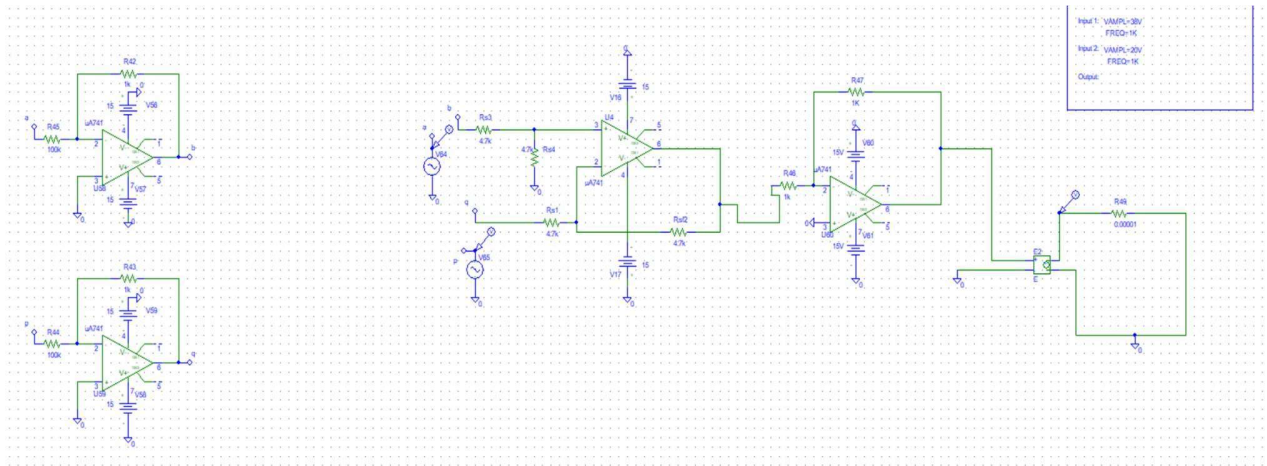


LIMITATION

If $\left| \frac{\text{input } 1}{100} \right| \geq 15$, $\left| \frac{\text{input } 2}{100} \right| \geq 15$ or $\left| \frac{\text{input } 1}{100} + \frac{\text{input } 2}{100} \right| \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Additionally, there are instances where the output may include decimal points when it should ideally yield integer results. These challenges represent critical areas of concern in our design.

SUBTRACTION:

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit facilitates the subtraction of two input signals. Both U58 and U59 serve as inverting amplifiers, effectively dividing the input signal by 100. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

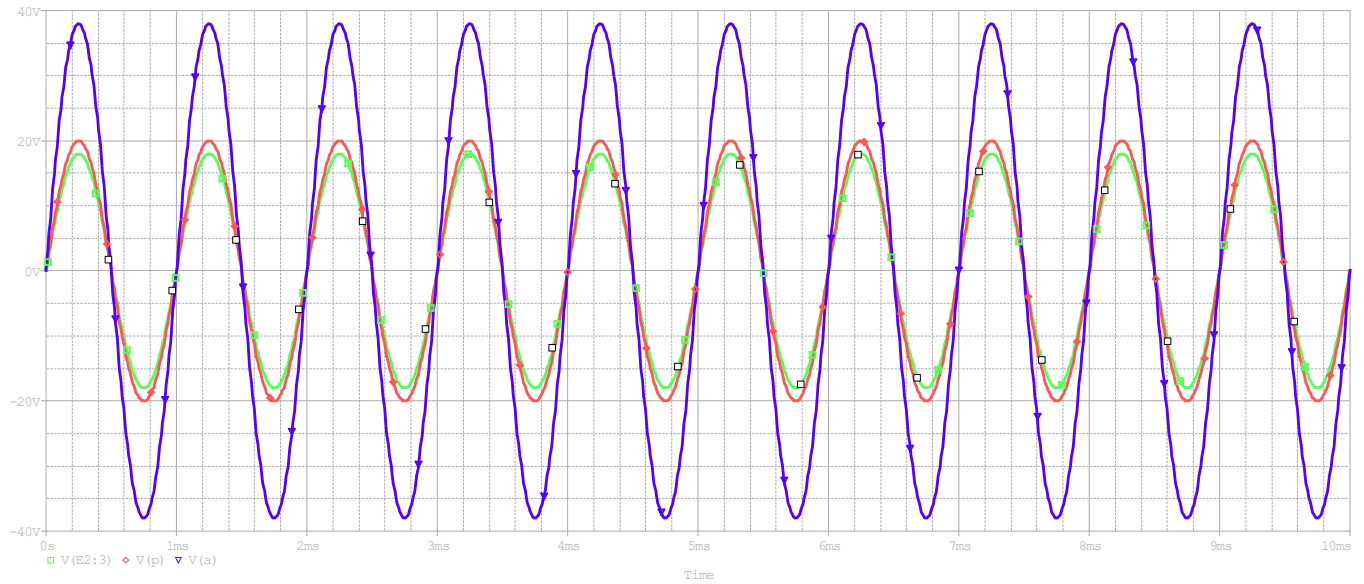
The U4 op amp works as a differential amplifier, subtracts non-inverting input from inverting input & produces the output signal. Subsequently, this output signal undergoes an inverting amplifier and a substantial 100-fold amplification through a voltage-controlled voltage source, ultimately restoring it to its original form.

Output:

Case 1:

Input 1: 38V p Sine wave, 1kHz

Input 2: 20V p Sine wave, 1kHz

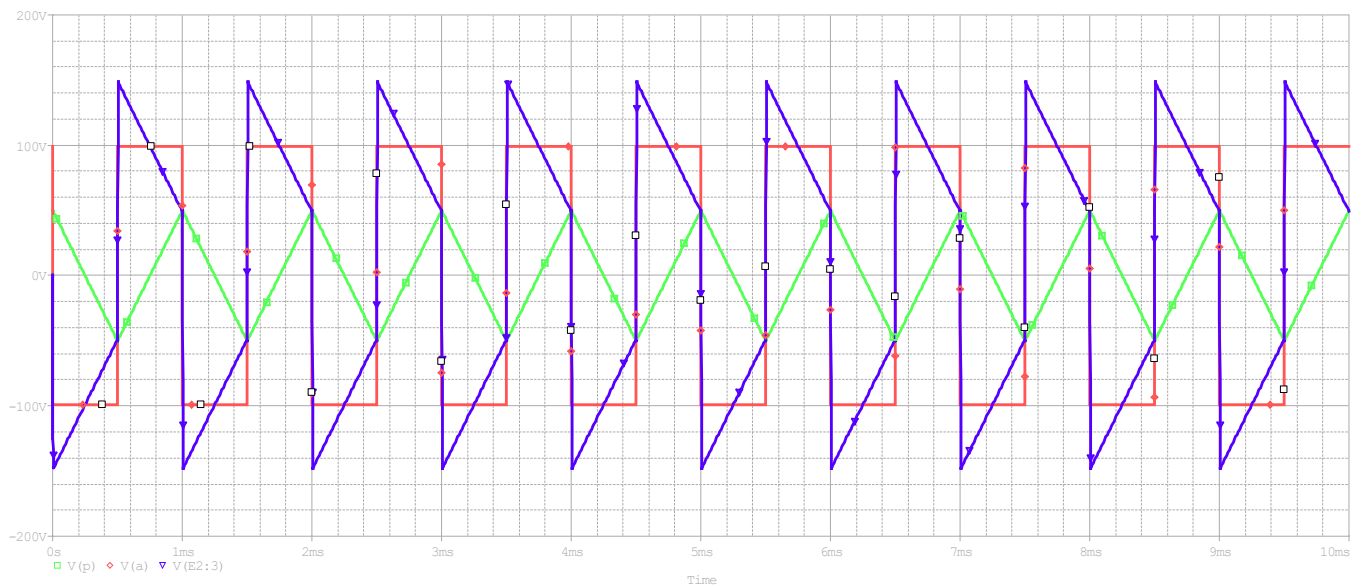


Here, blue and red are the input & green is output graph.

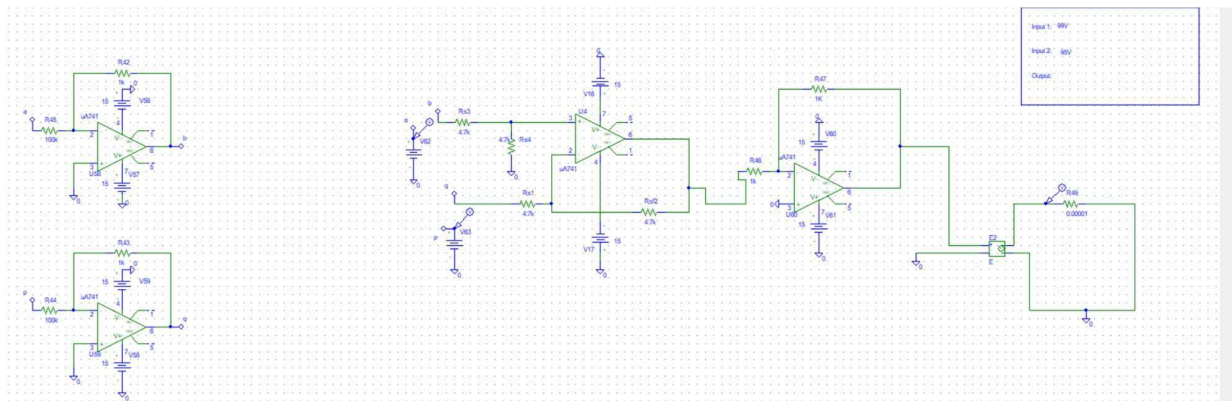
Case 2:

Input 1: 99 V Square wave, 1kHz &

Input 2: 50V Triangular wave, 1kHz



CIRCUIT & INPUT: (DC)



Output:

Case 1:

Input 1: 99V

Input 2: 95V

Output: 4.005V

Case 2:

Input 1: 50V

Input 2: 95V

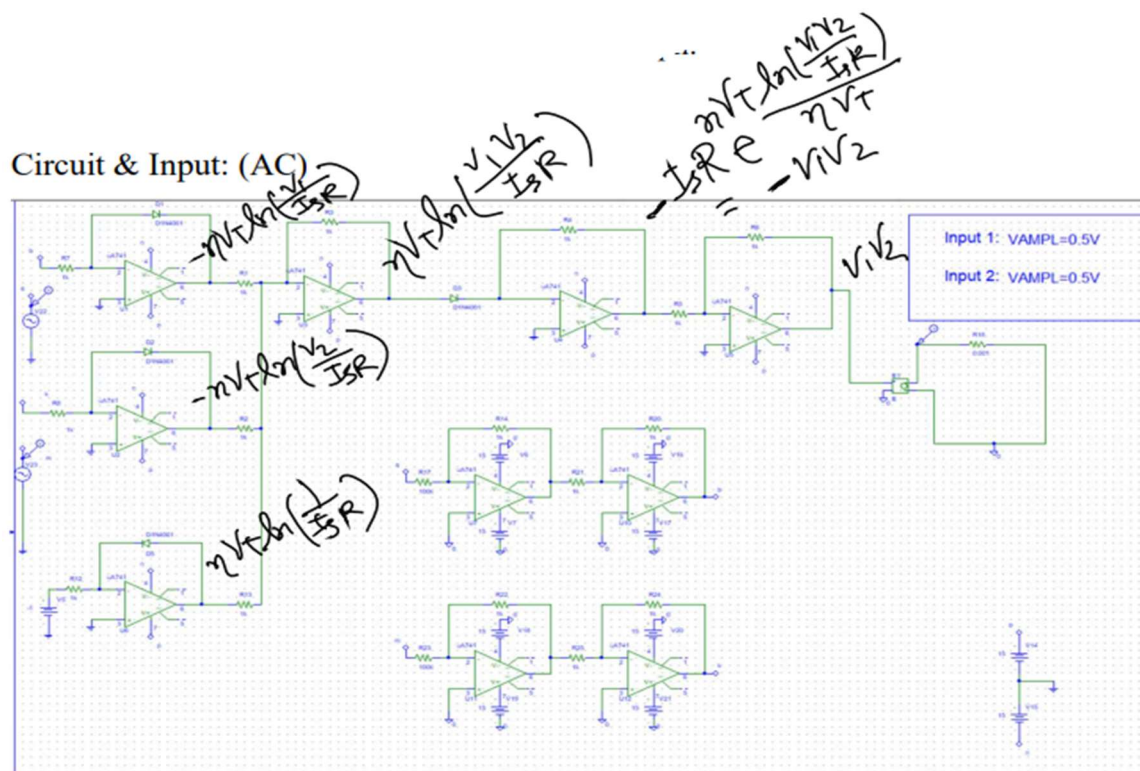
Output: -44.99V

LIMITATION

If $\left| \frac{\text{input 1}}{100} \right| \geq 15$, $\left| \frac{\text{input 2}}{100} \right| \geq 15$ or $\left| \frac{\text{input 1}}{100} - \frac{\text{input 2}}{100} \right| \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Additionally, there are instances where the output may include decimal points when it should ideally yield integer results. These challenges represent critical areas of concern in our design.

MULTIPLIER:

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit is designed to perform the multiplication of two input signals with precision and versatility. It utilizes operational amplifiers to ensure accurate and controlled signal processing.

For the first input signal (Input 1), U7 initially divides it by 100, and then U10 further processes it by serving as an inverting amplifier. The same process is applied to Input 2. This careful division and inversion of the input signals are essential in accommodating 2-digit input values while effectively preventing the operational amplifiers from reaching saturation.

Beyond these initial stages, the circuit incorporates a set of inverting logarithmic amplifiers for both input voltages and a non-inverting logarithmic amplifier for a -1 volt signal. These three processed voltages are then combined within an inverting adder and subsequently pass through an inverting exponential amplifier. Following this, the signal undergoes further refinement through an inverting amplifier.

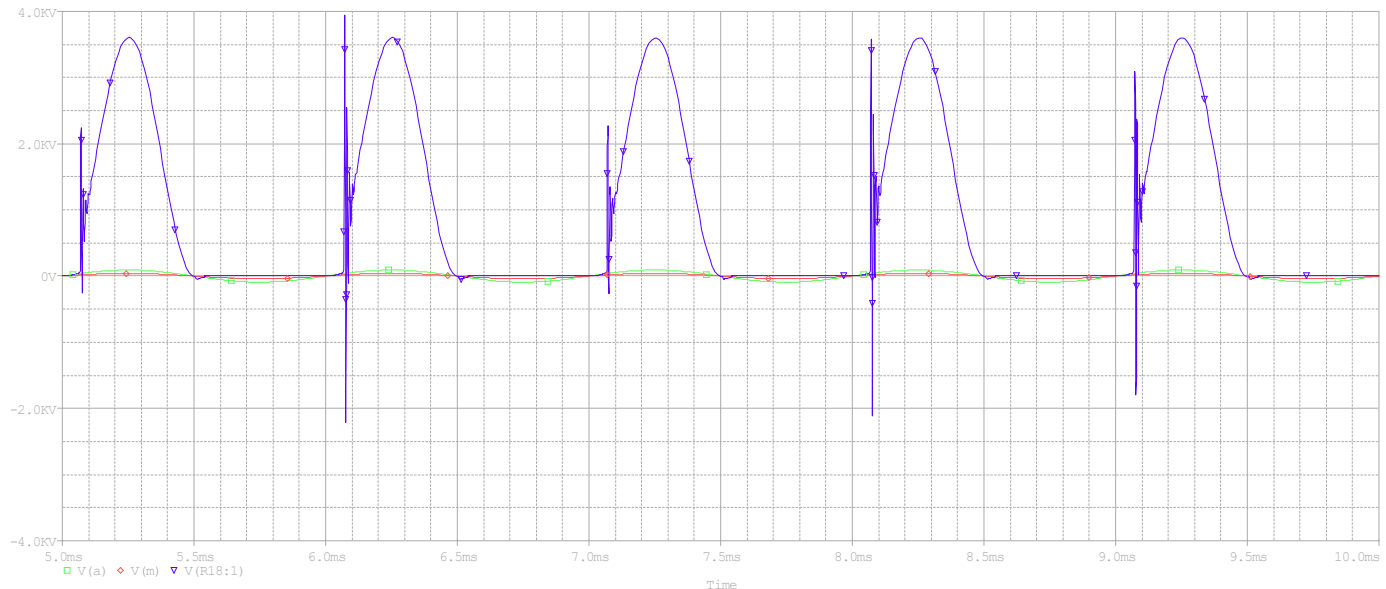
One of the critical aspects of this design is the substantial 10,000-fold amplification applied to the output signal using a voltage-controlled voltage source. This amplification process ensures that the final output signal is adequately scaled, effectively restoring it to its original form.

Output:

Case 1:

Input 1: 90V p Sine wave, 1kHz

Input 2: 40V p Sine wave, 1kHz

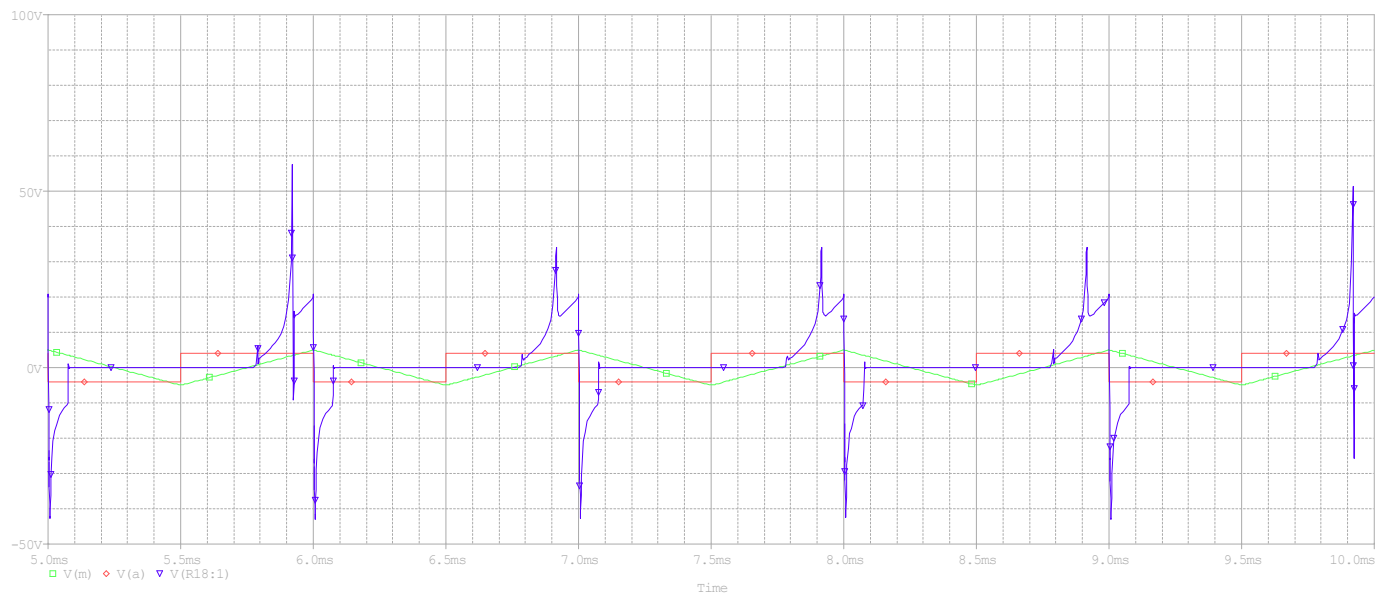


Here, green and red are the input & blue is output graph.

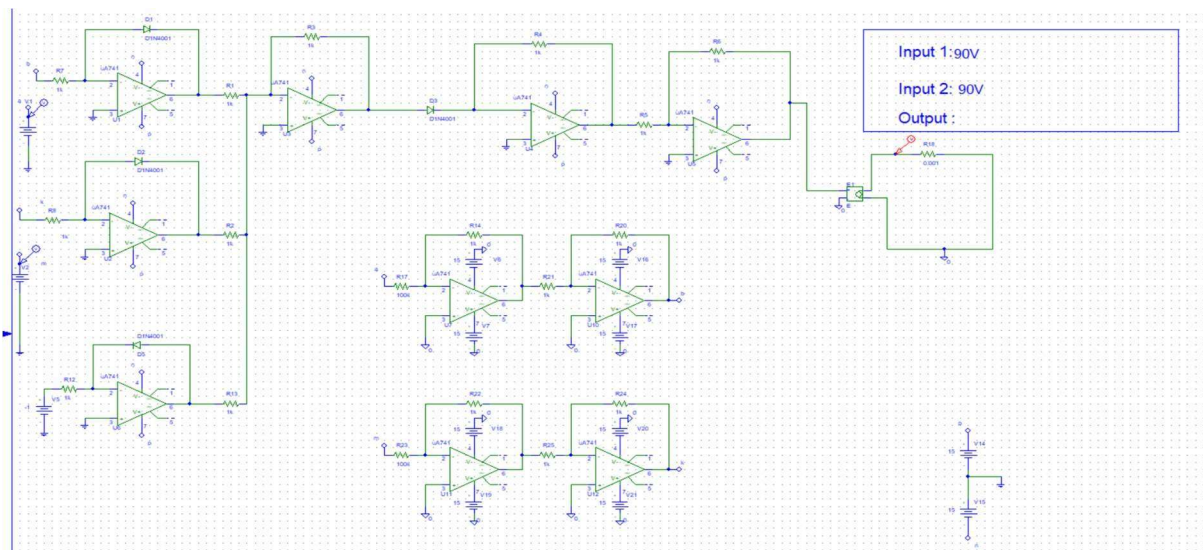
Case 2:

Input 1: 4 V Square wave, 1kHz &

Input 2: 5V Triangular wave, 1kHz



CIRCUIT & INPUT: (DC)



Output:

Case 1:

Input 1: 90V
Input 2: 90V
Output : 8.106KV

Case 2:

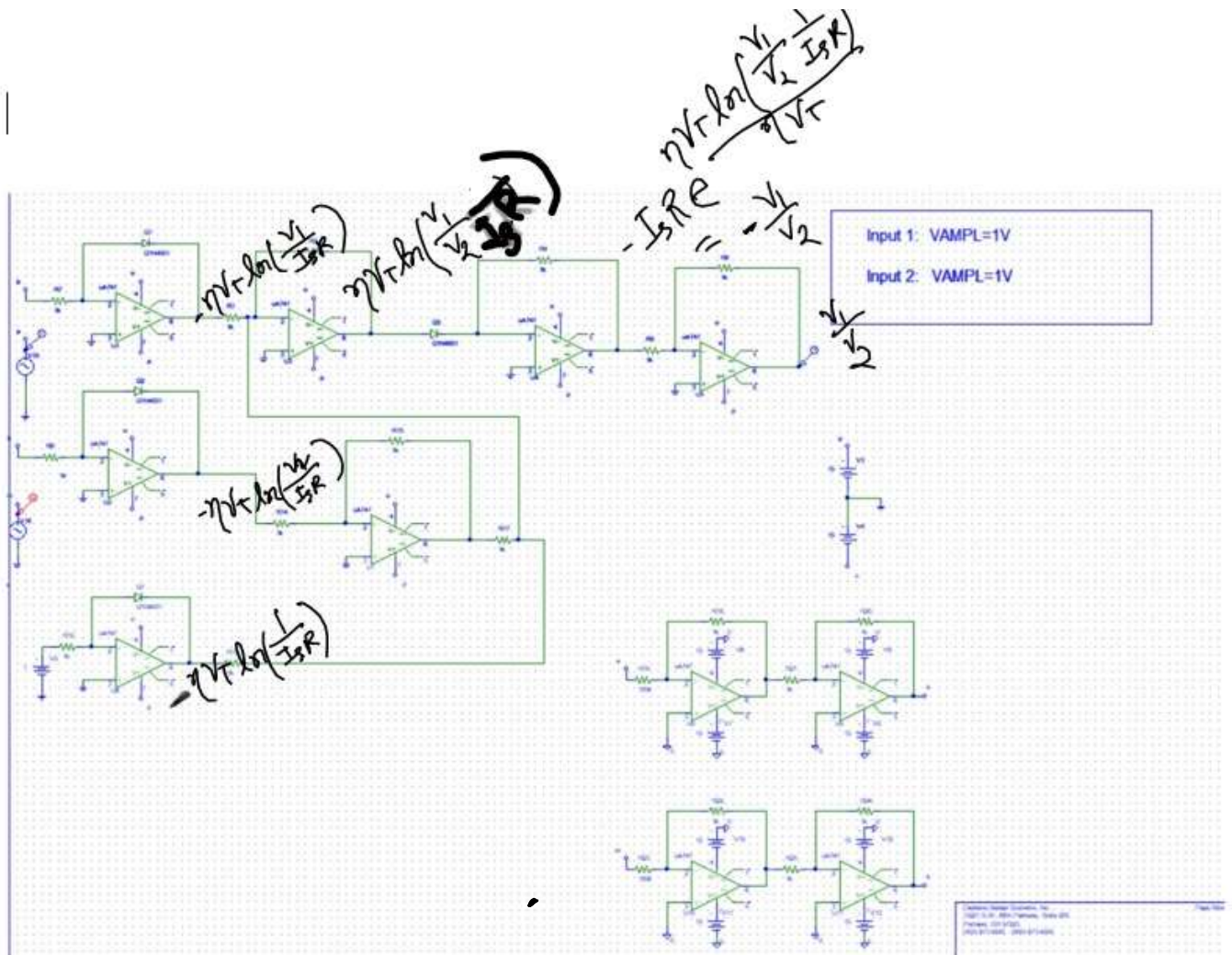
Input 1: 80V
Input 2: 90V
Output : 7.206KV

LIMITATION

If $\left| \frac{\text{input } 1}{100} \right| \geq 15$, $\left| \frac{\text{input } 2}{100} \right| \geq 15$ or $\left| \frac{\text{input } 1}{100} * \frac{\text{input } 2}{100} \right| \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Additionally, there are instances where the output may include decimal points when it should ideally yield integer results. These challenges represent critical areas of concern in our design. Moreover, A voltage-controlled voltage source (VCVS) with a gain of 10,000 is theoretically possible, but it would be challenging to implement directly to practical limitations

DIVIDER:

CIRCUIT & INPUT: (AC)



EXPLANATION

This circuit is designed to perform the division of two input signals with precision and versatility. It utilizes operational amplifiers to ensure accurate and controlled signal processing.

For the first input signal (Input 1), U7 initially divides it by 100, and then U10 further processes it by serving as an inverting amplifier. The same process is applied to Input 2. This careful division and inversion of the input signals are essential in accommodating 2-digit input values while effectively preventing the operational amplifiers from reaching saturation.

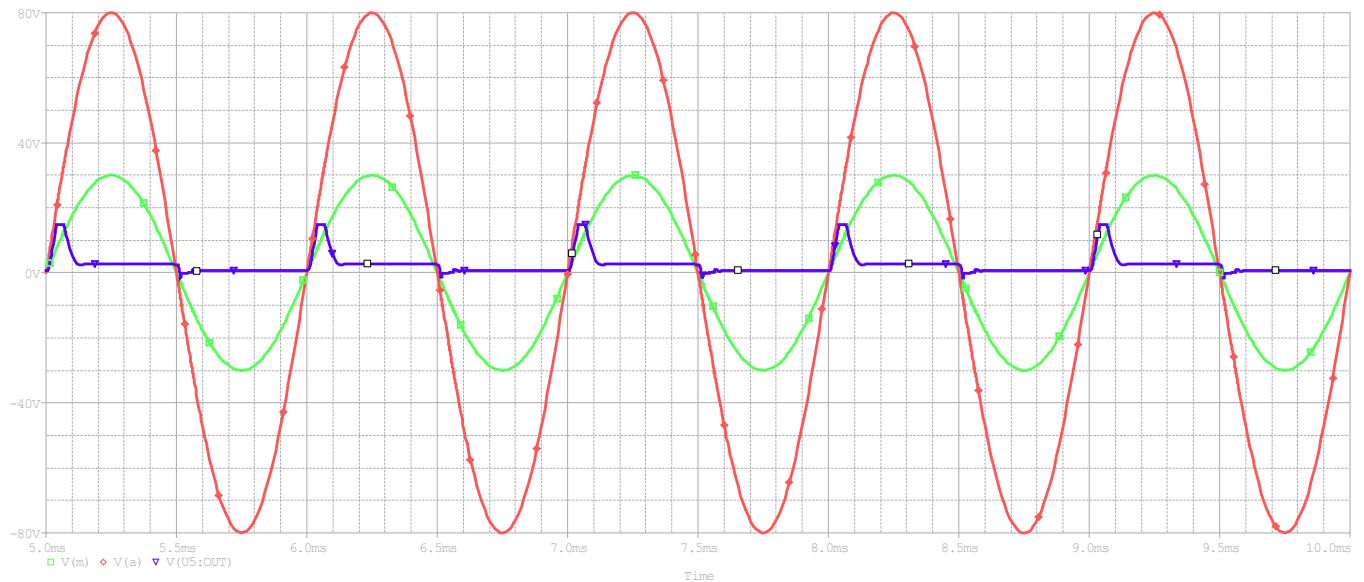
Beyond these initial stages, the circuit incorporates a set of inverting logarithmic amplifiers for both input voltages and an inverting logarithmic amplifier for a 1 volt signal. These three processed voltages are then combined within an inverting adder and subsequently pass through an inverting exponential amplifier. Following this, the signal undergoes further refinement through an inverting amplifier.

Output:

Case 1:

Input 1: 80V p Sine wave, 1kHz

Input 2: 30V p Sine wave, 1kHz

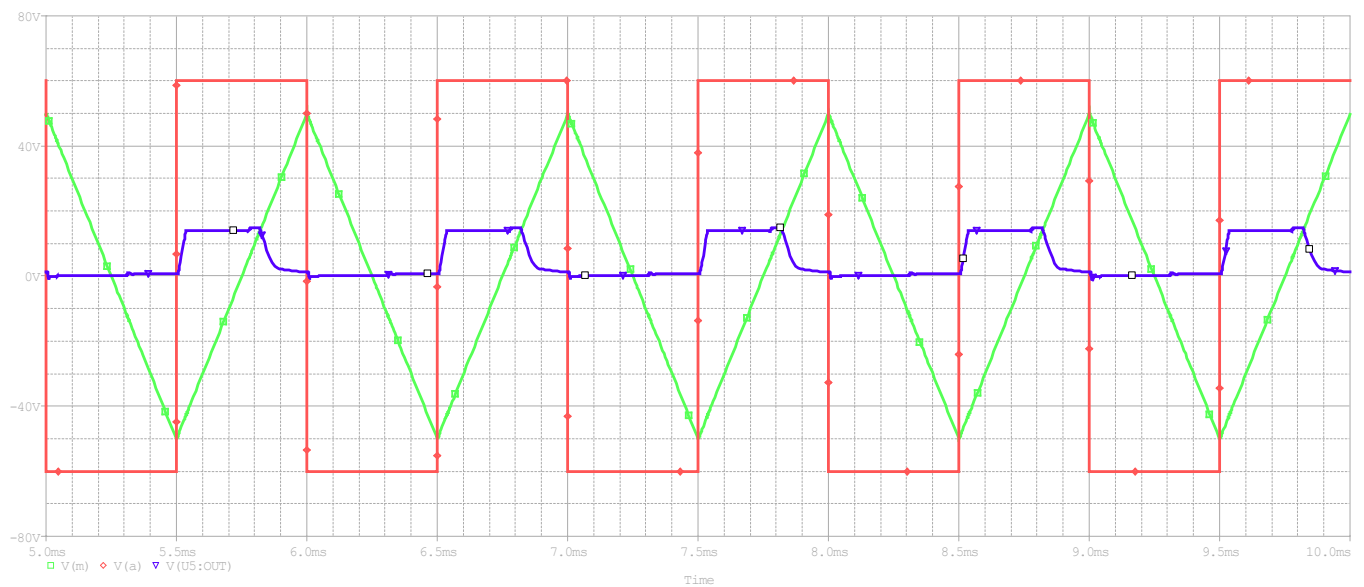


Here, green and red are the input & blue is output graph.

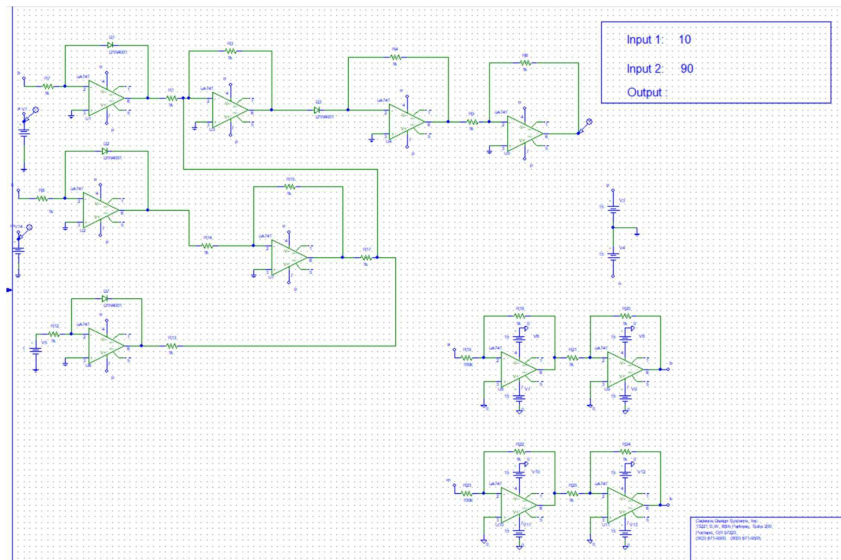
Case 2:

Input 1: 60 V Square wave, 1kHz &

Input 2: 50V Triangular wave, 1kHz

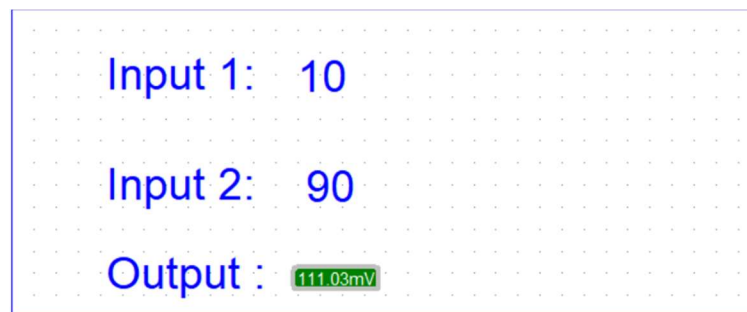


CIRCUIT & INPUT: (DC)

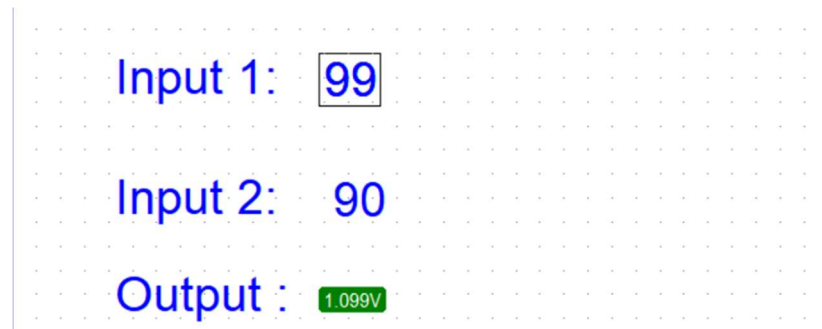


Output:

Case 1:



Case 2:

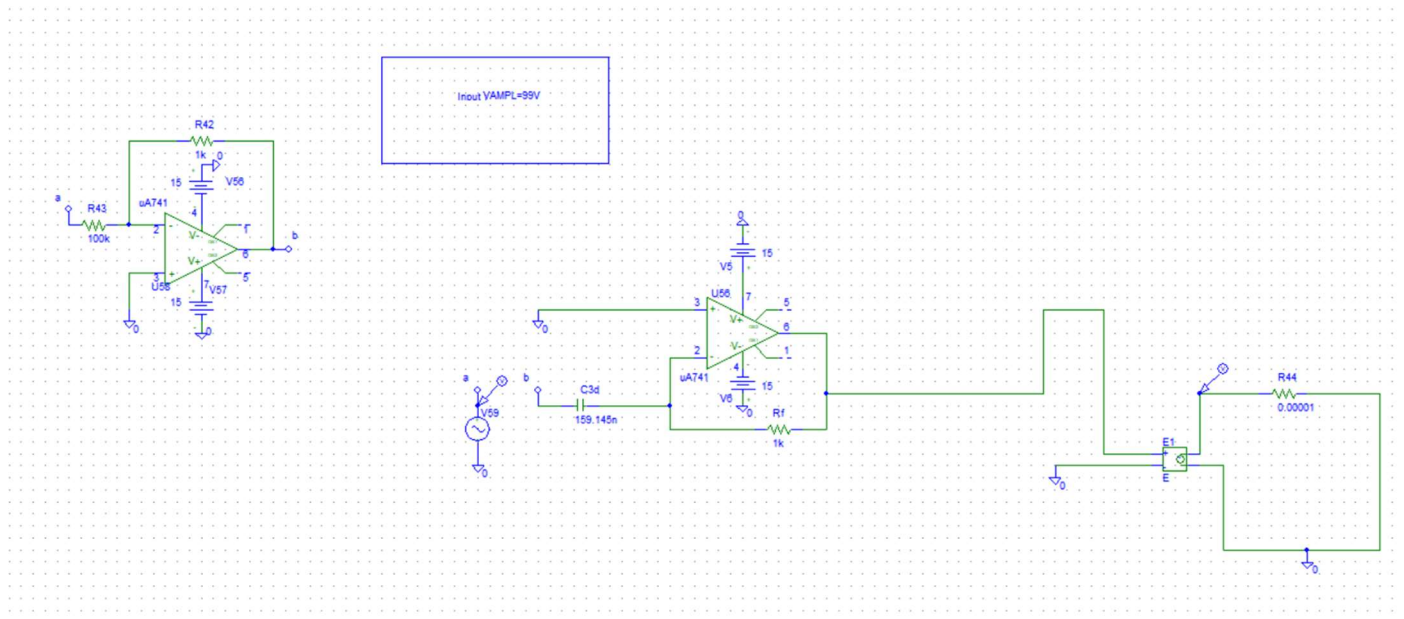


LIMITATION

If $\left| \frac{\text{input } 1}{100} \right| \geq 15$, $\left| \frac{\text{input } 2}{100} \right| \geq 15$ or $\left| \frac{\text{input } 1}{100} / \frac{\text{input } 2}{100} \right| \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Additionally, there are instances where the output may include decimal points when it should ideally yield integer results. These challenges represent critical areas of concern in our design.

DIFFERENTIATOR:

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit facilitates the differentiation of input signal. U58 serves as inverting amplifiers, effectively dividing the input signal by 100. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

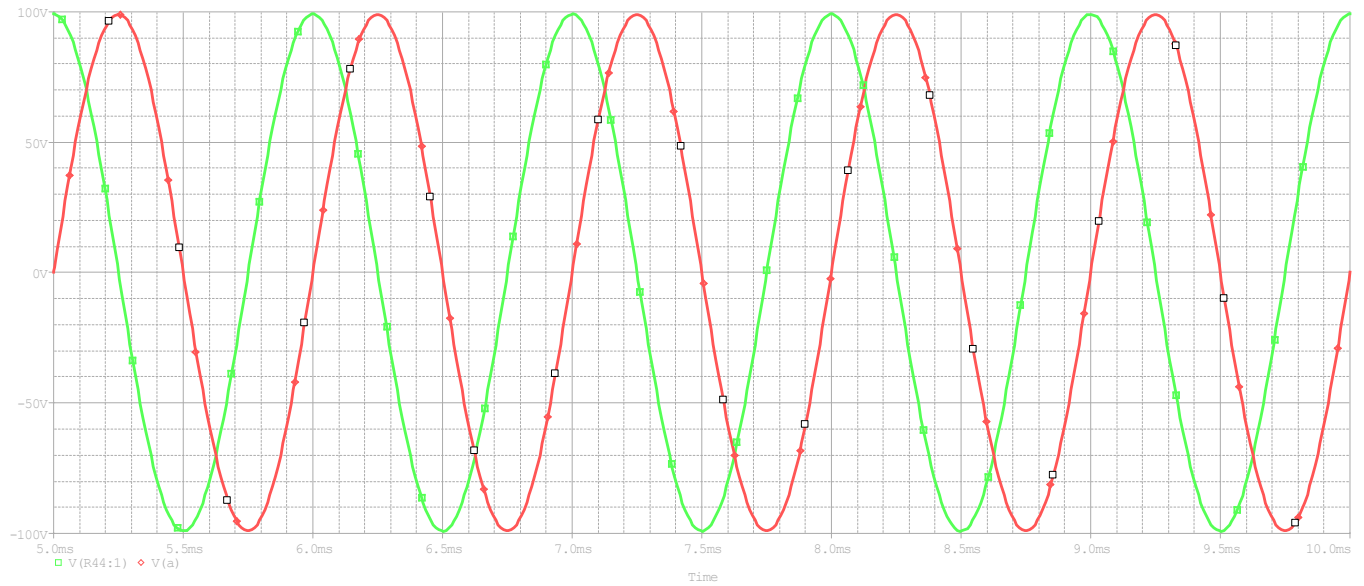
Operational amplifiers U56 and U57 play crucial roles in shaping the output signal. U56 operates as an inverting differentiator which generates the desired output signal, refining it to meet our specifications.

Following this stage, the output signal undergoes a substantial 100-fold amplification, a process expertly executed by a voltage-controlled voltage source. This amplification step serves to magnify the signal significantly, ultimately restoring it to its original, desired form.

Output:

Case 1:

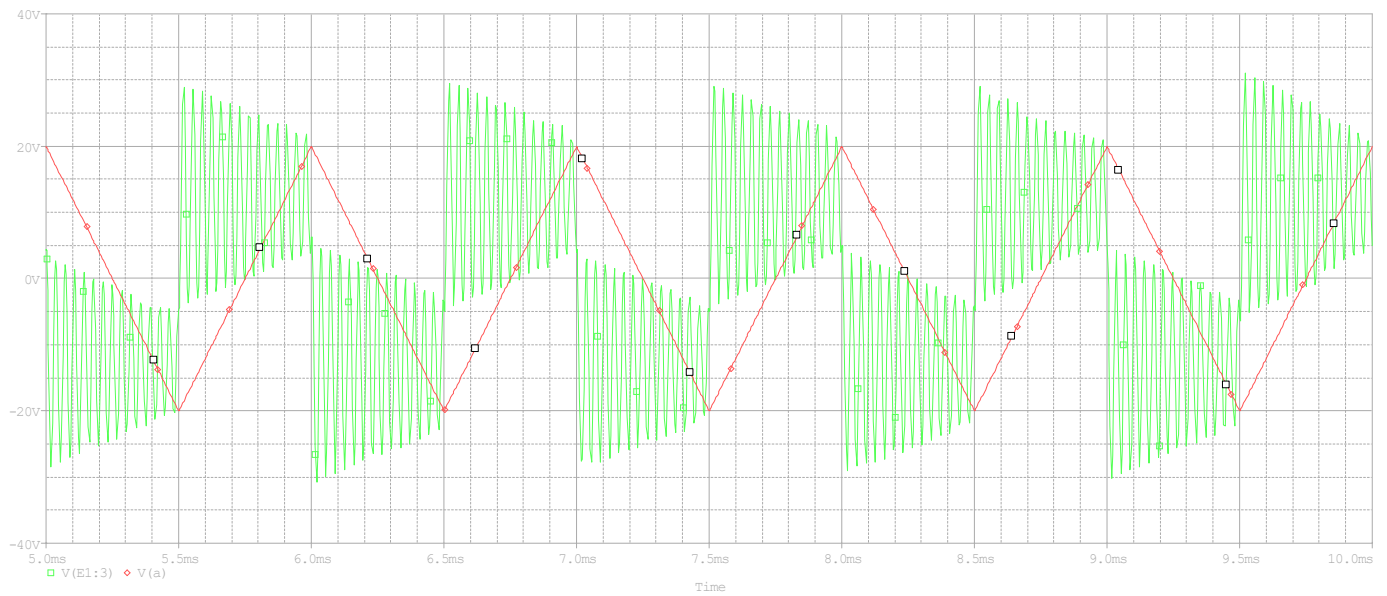
Input 1: 99V p Sine wave, 1kHz



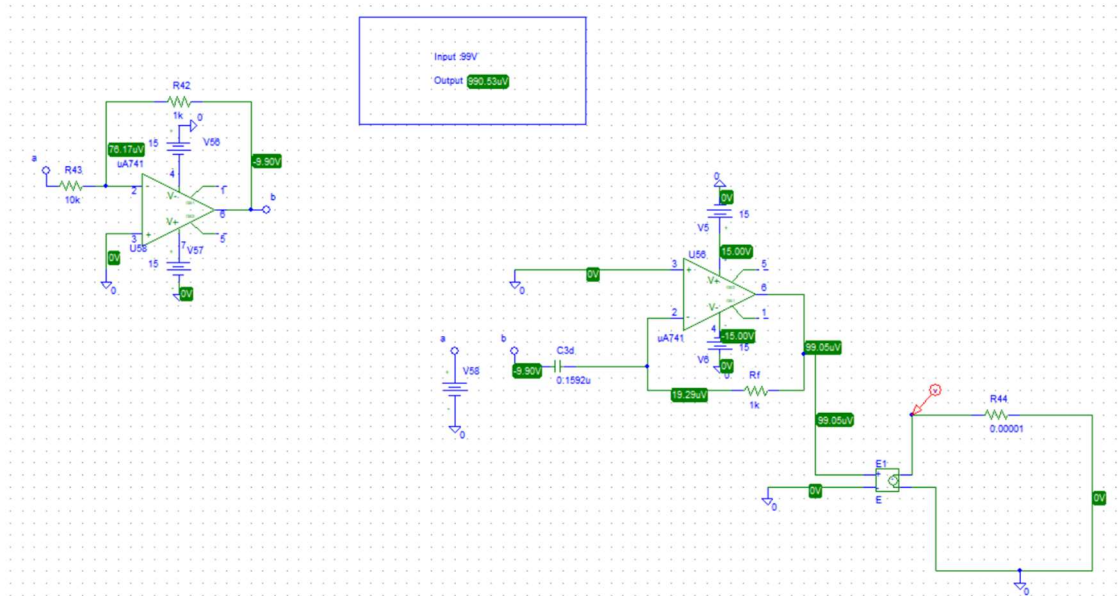
Here, red is the input & green is output graph.

Case 2:

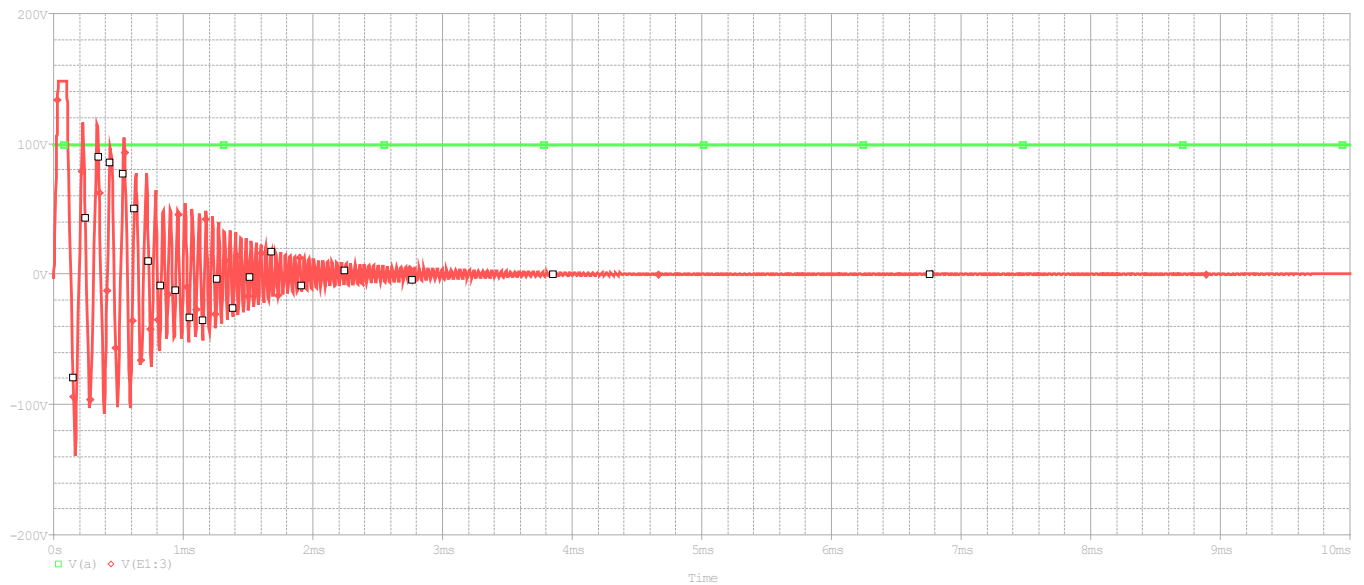
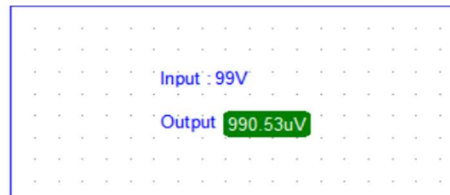
Input 1: 20V p Triangular wave, 1kHz



CIRCUIT & INPUT: (DC)



Output:



Here, red is the output & green is input graph.

While it's expected that the differentiation of a constant value (in this case, 99) should ideally result in zero, our circuit exhibits a small margin of error. Specifically, the output registers at 990.53 microvolts, which is remarkably close to the expected zero value. However, this minor deviation warrants attention and optimization to further minimize any residual error in our output.

LIMITATION

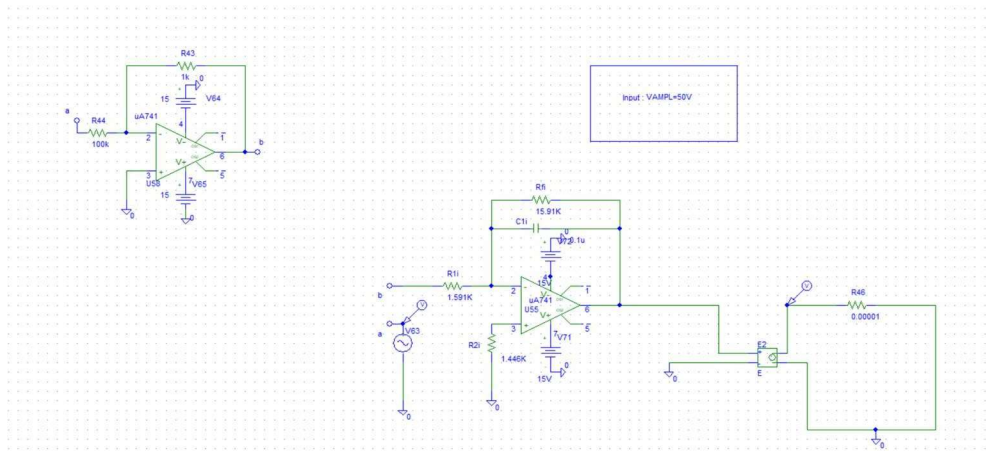
If $\left| \frac{\text{input } 1}{100} \right| \geq 15$ or $RC \frac{d}{dt} \left| \frac{\text{input } 1}{100} \right| \geq 15 V$ then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Furthermore, it's noteworthy that our circuit may occasionally produce output values with decimal points, whereas integer results are desired. This particular issue stands out as a critical concern in our design, calling for meticulous attention and optimization to ensure precise integer outcomes.

In the case of our AC differential circuit, it's imperative to acknowledge that it is optimized for input signals with a 1 kHz frequency. When the input signal deviates from this frequency, adjustments to the value of the capacitor (C) are required to maintain a gain of 1. Consequently, our AC circuit's applicability is restricted to input signals with a 1 kHz frequency profile.

In contrast, our DC differential circuit exhibits an inherent characteristic of not zeroing out at direct current (DC) input; instead, it provides a value that approximates zero volts. This subtle deviation necessitates our continued focus on refinement to further reduce any residual error in the output signal.

INTEGRATOR:

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit facilitates the integration of input signal. U58 serves as inverting amplifiers, effectively dividing the input signal by 100. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

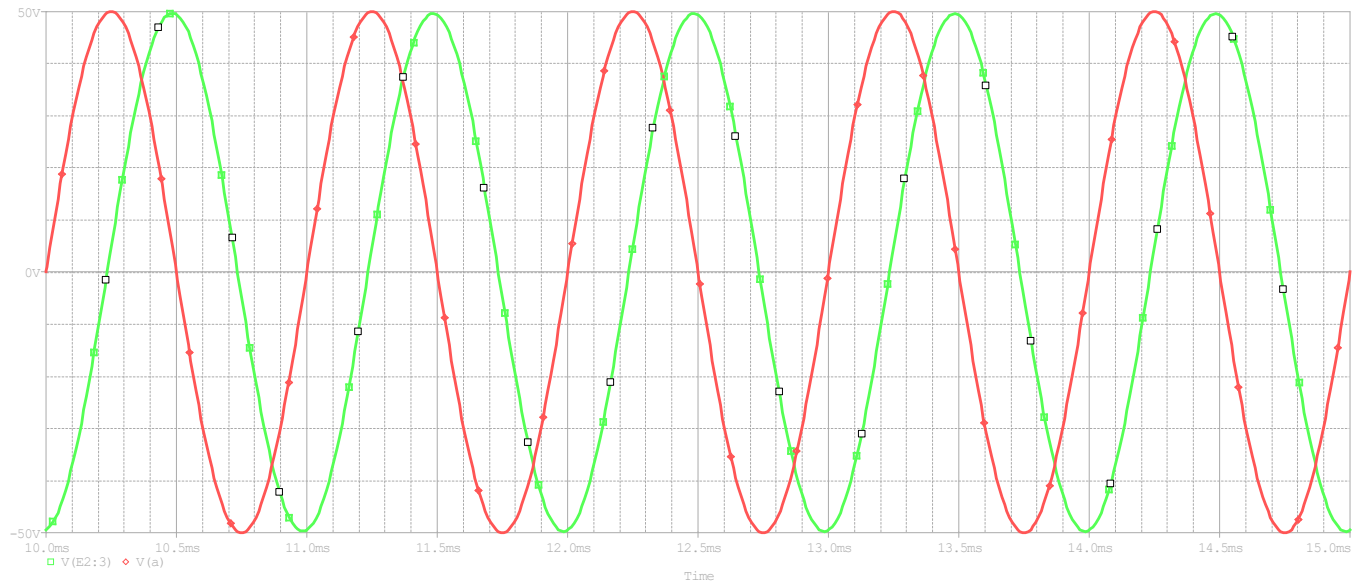
Operational amplifiers U55 plays crucial roles in shaping the output signal. U55 operates as a practical inverting integrator which generates the desired output signal, refining it to meet our specifications.

Following this stage, the output signal undergoes a substantial 100-fold amplification, a process expertly executed by a voltage-controlled voltage source. This amplification step serves to magnify the signal significantly, ultimately restoring it to its original, desired form.

Output:

Case 1:

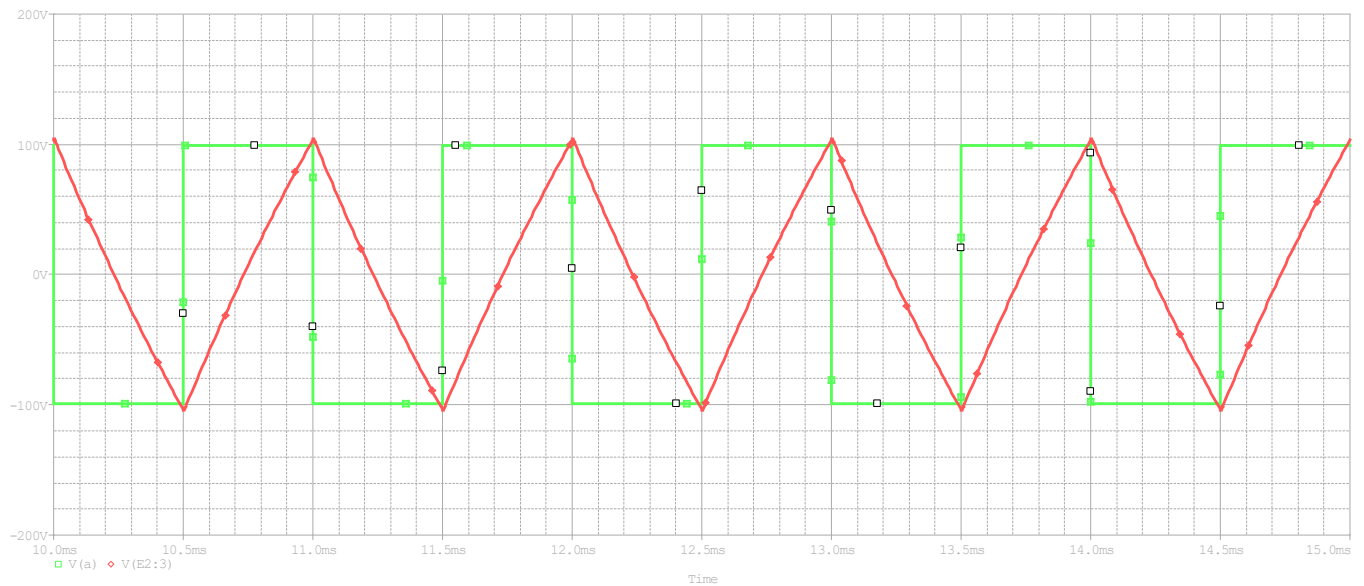
Input 1: 50V p Sine wave, 1kHz



Here, red is the input & green is output graph.

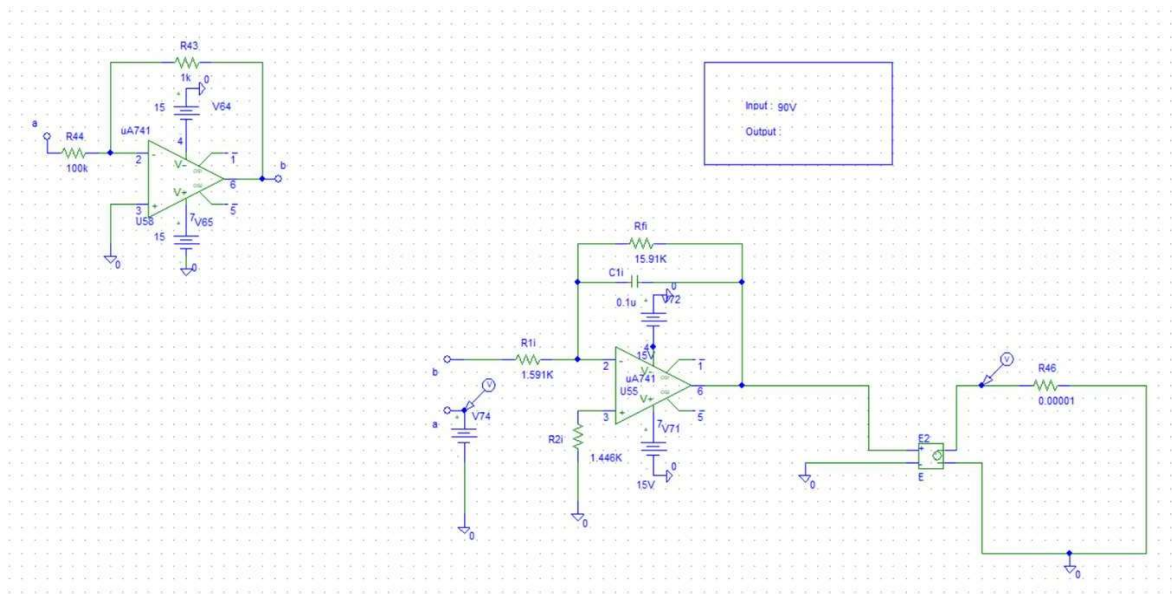
Case 2:

Input 1: 99V p square wave, 1kHz

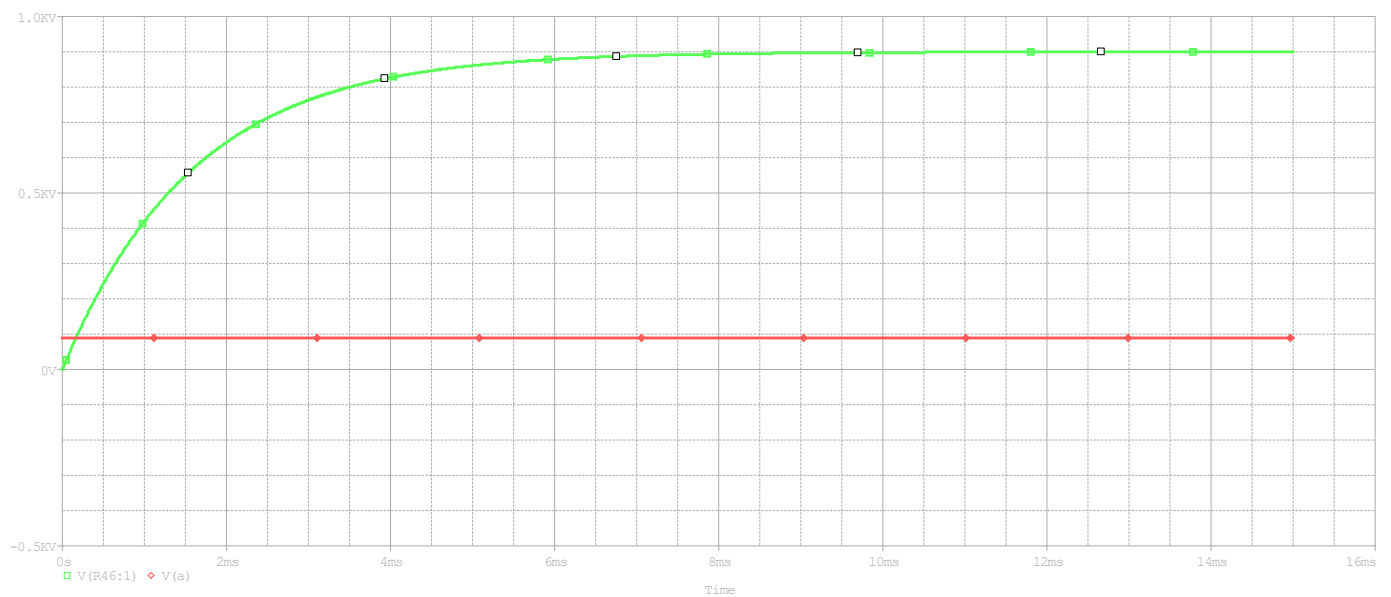
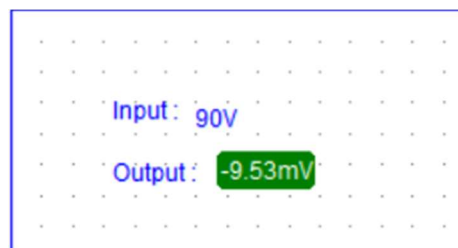


Here, red is the output & green is input graph.

CIRCUIT & INPUT: (DC)



Output:



Here, red is the input & green is output graph.

LIMITATION

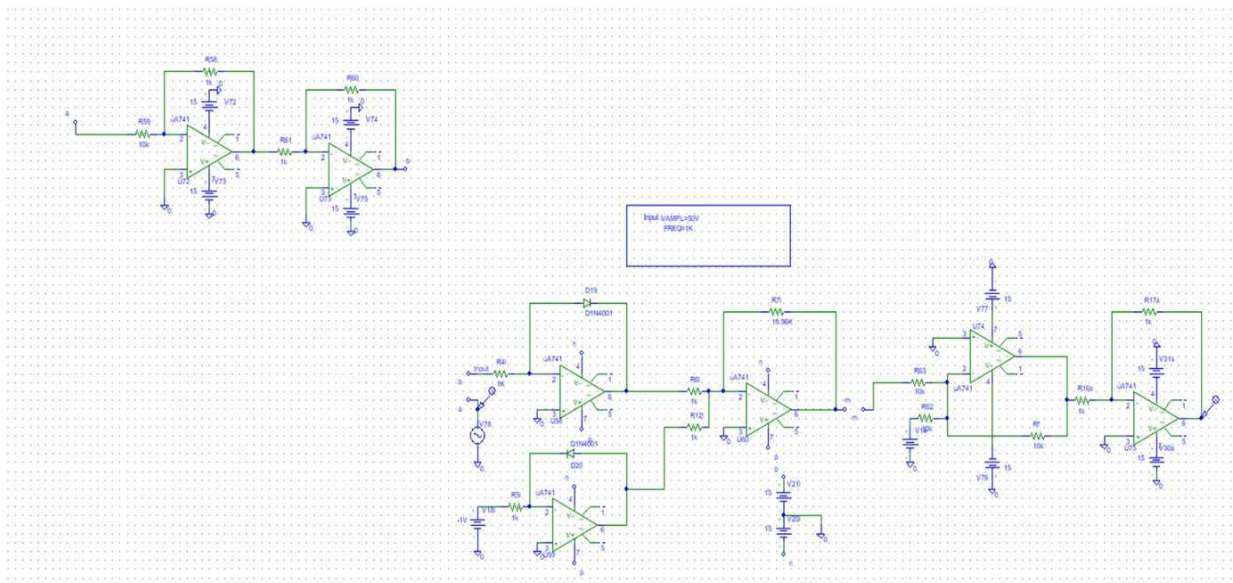
If $\left| \frac{\text{input } 1}{100} \right| \geq 15$ or $\frac{1}{RC} \int \left| \frac{\text{input } 1}{100} \right| dt \geq 15 V$ then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. Furthermore, it's noteworthy that our circuit may occasionally produce output values with decimal points, whereas integer results are desired. This particular issue stands out as a critical concern in our design, calling for meticulous attention and optimization to ensure precise integer outcomes.

In the case of our AC integrating circuit, it's imperative to acknowledge that it is optimized for input signals with a 1 kHz frequency. When the input signal deviates from this frequency, adjustments to the value of the capacitor (C) are required to maintain a gain of 1. Consequently, our AC circuit's applicability is restricted to input signals with a 1 kHz frequency profile.

In contrast, our DC integrating circuit displays a distinctive behavior when subjected to direct current (DC) input. Ideally, it should yield a linear relationship represented by a $y = mx$ graph, but instead, it manifests as a parabolic graph that saturates at a relatively high value. This unexpected behavior underscores the importance of our ongoing efforts to refine the circuit design and minimize any lingering errors in the output signal.

NATURAL LOGARITHM:

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. This circuit provides the logarithm of input signal. U72 serves as inverting amplifiers, effectively dividing the input signal by 10 and U73 serves as an inverting amplifier making the signal positive again. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

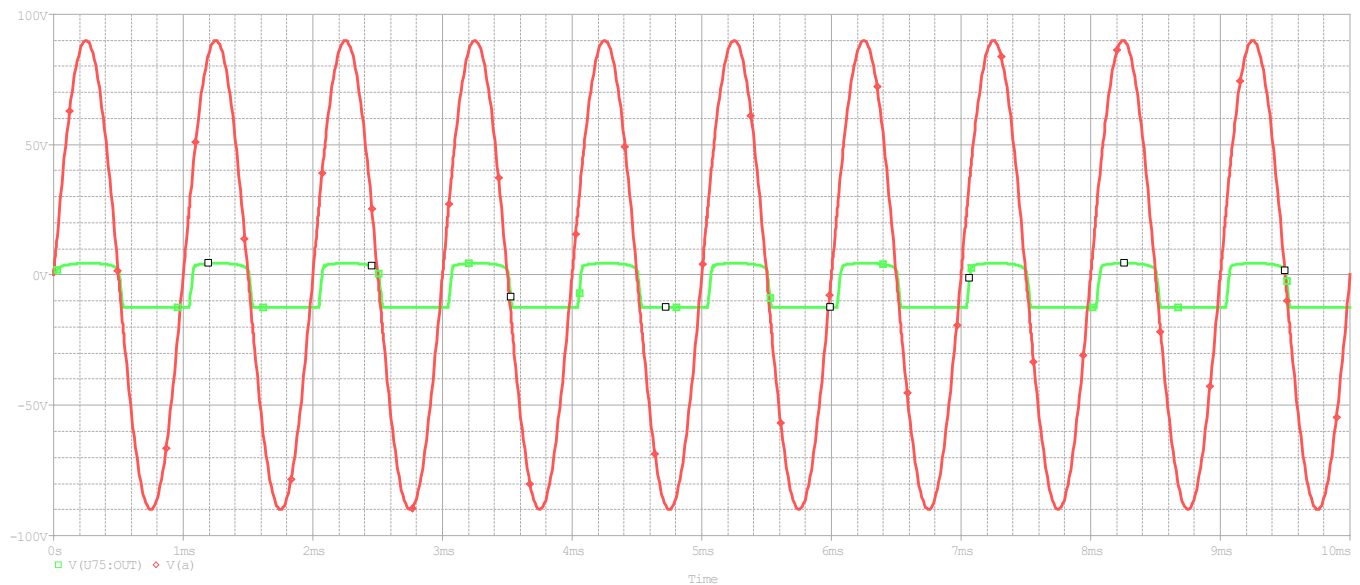
The process of obtaining the logarithmic output from the input signal involves passing it through U58. However, within this logarithmic function, a fractional term emerges, which needs to be nullified. This crucial nullification step is skillfully executed by U59, an operational amplifier dedicated to this task. Subsequently, U7, serving as another operational amplifier, adjusts the phase of the output, effectively counteracting the inversion and neutralizing any undesired coefficients within the logarithmic output.

The journey of our signal continues as it traverses an inverting summer, where it is combined with a signal of $\ln(10)$ amplitude. This step is instrumental in restoring the signal to its original state. Finally, U75, operating as an inverting amplifier, meticulously refines the output signal to align with our specified requirements and criteria.

Output:

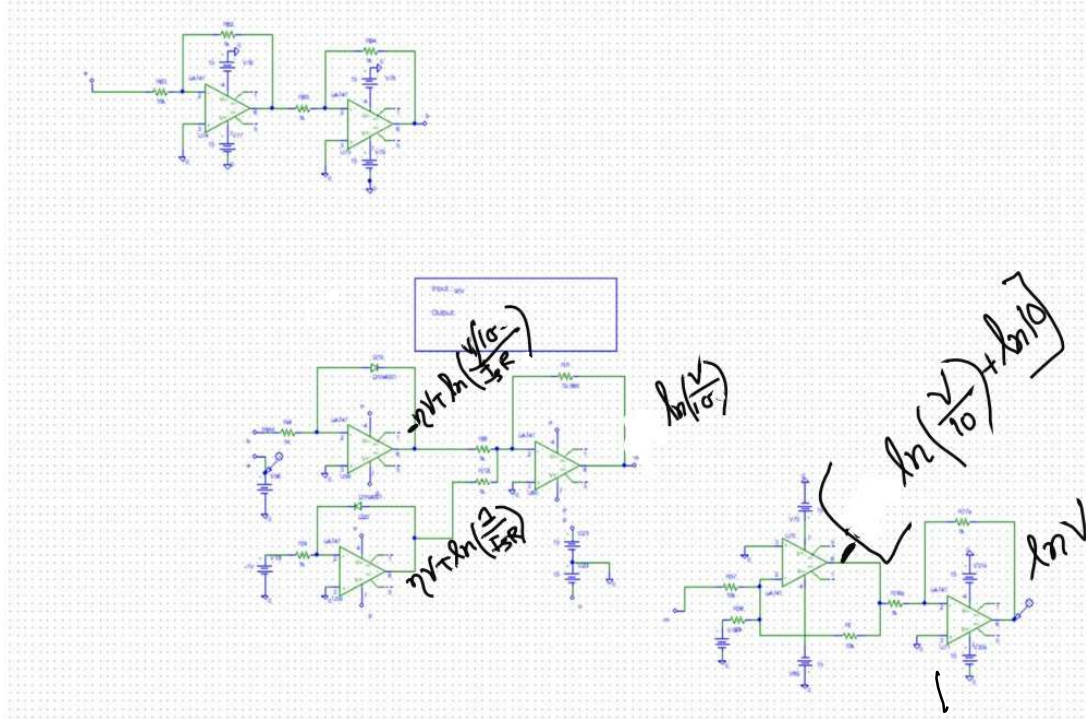
Case 1:

Input 1: 90V p Sine wave, 1kHz

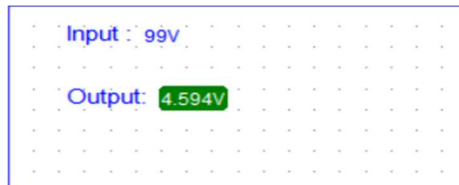


Here, red is the input & green is output graph.

CIRCUIT & INPUT: (DC)



Output:

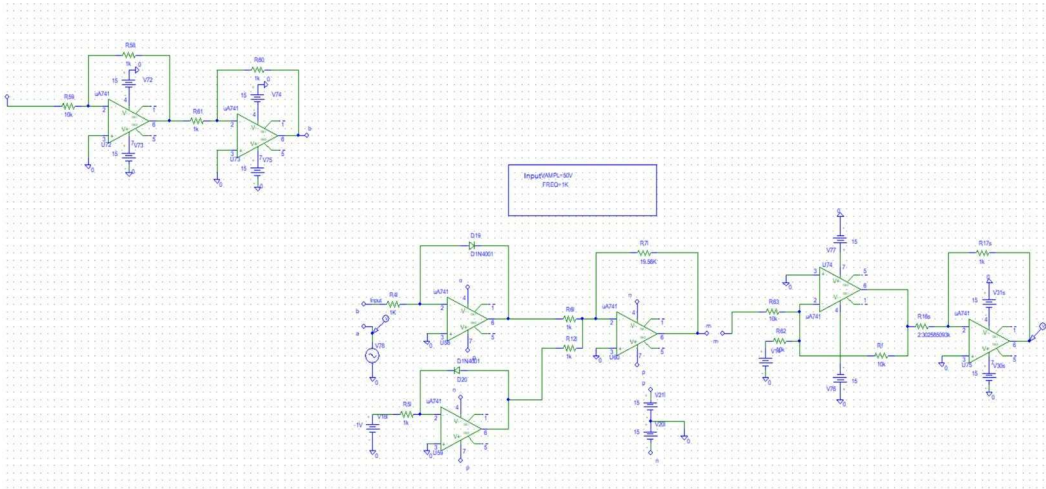


LIMITATION

If $\left| \frac{\text{input } 1}{10} \right| \geq 15$ or $\ln(\text{input } 1) \geq 15 \text{ V}$ then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. At times, our circuit may yield decimal values that do not perfectly align with the results expected from a real-life calculator. This particular issue is of paramount importance and warrants rigorous attention and optimization to ensure that our circuit consistently produces precise outcomes that match the expectations of real-life calculations.

LOGARITHM (BASE 10):

CIRCUIT & INPUT: (AC)



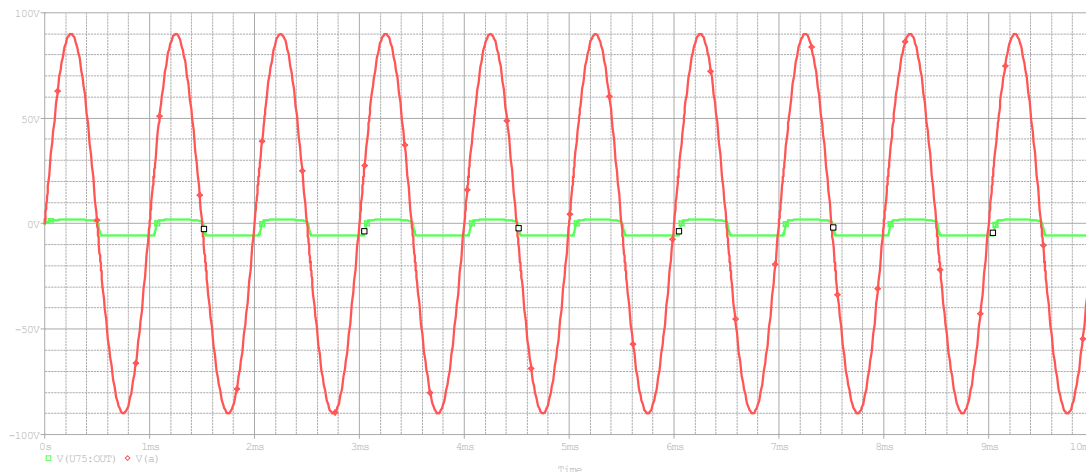
EXPLANATION

We know, $\log_{10} X = \frac{\ln(X)}{2.303}$. By introducing a slight modification to our natural logarithm calculation circuit, we can extend its functionality to calculate logarithms with a base of 10. This adjustment is implemented within U75, where we change the value of input resistor R to 2.303k. As a result, the output is effectively divided by 2.303, allowing us to obtain logarithmic values with a base of 10. This enhancement broadens the utility and versatility of our circuit for logarithmic calculations.

Output:

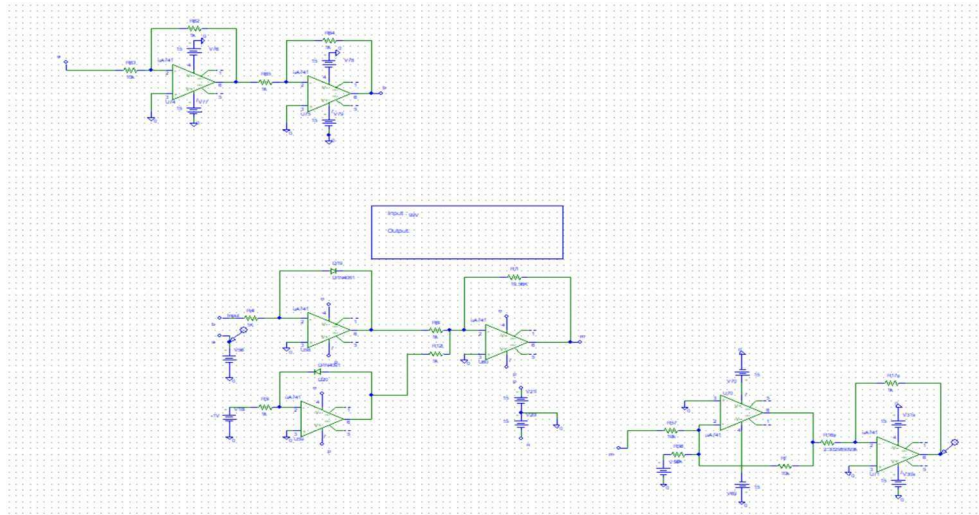
Case 1:

Input 1: 90V p Sine wave, 1kHz



Here, red is the input & green is output graph.

CIRCUIT & INPUT: (DC)



Output:

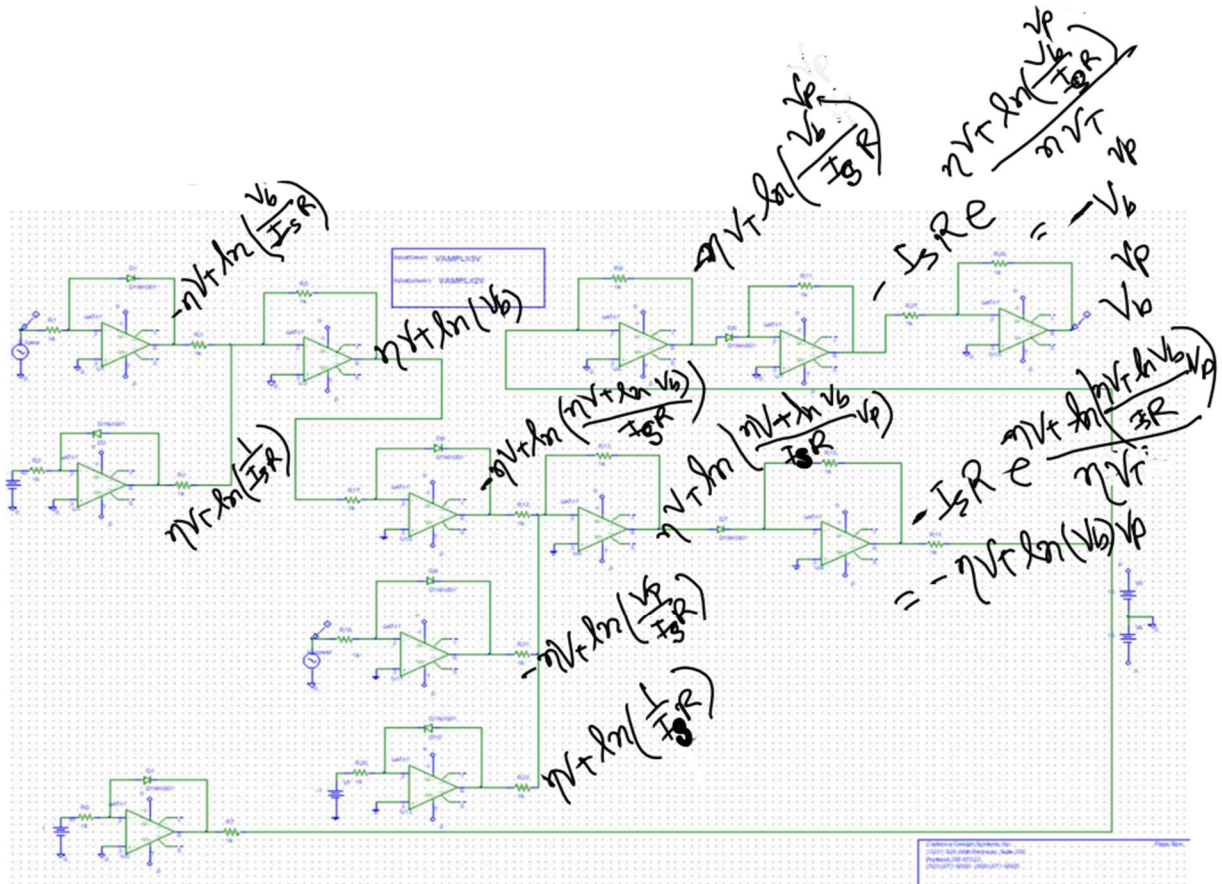


LIMITATION

If $\left| \frac{\text{input } 1}{10} \right| \geq 15$ or $\log_{10}(\text{input } 1) \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. At times, our circuit may yield decimal values that do not perfectly align with the results expected from a real-life calculator. This particular issue is of paramount importance and warrants rigorous attention and optimization to ensure that our circuit consistently produces precise outcomes that match the expectations of real-life calculations.

POWER

CIRCUIT & INPUT: (AC)



EXPLANATION

Here bias voltage is 15 V and -15 V. For a more detailed understanding, please refer to the accompanying schematic diagram, which elucidates the internal procedures with the relevant equations.

Operational amplifier U1, functioning as a logarithmic amplifier, plays a pivotal role in introducing the input signal, effectively serving as the base in our process. Simultaneously, U4, another operational amplifier, is employed to neutralize any undesired factors that manifest within the logarithmic equation.

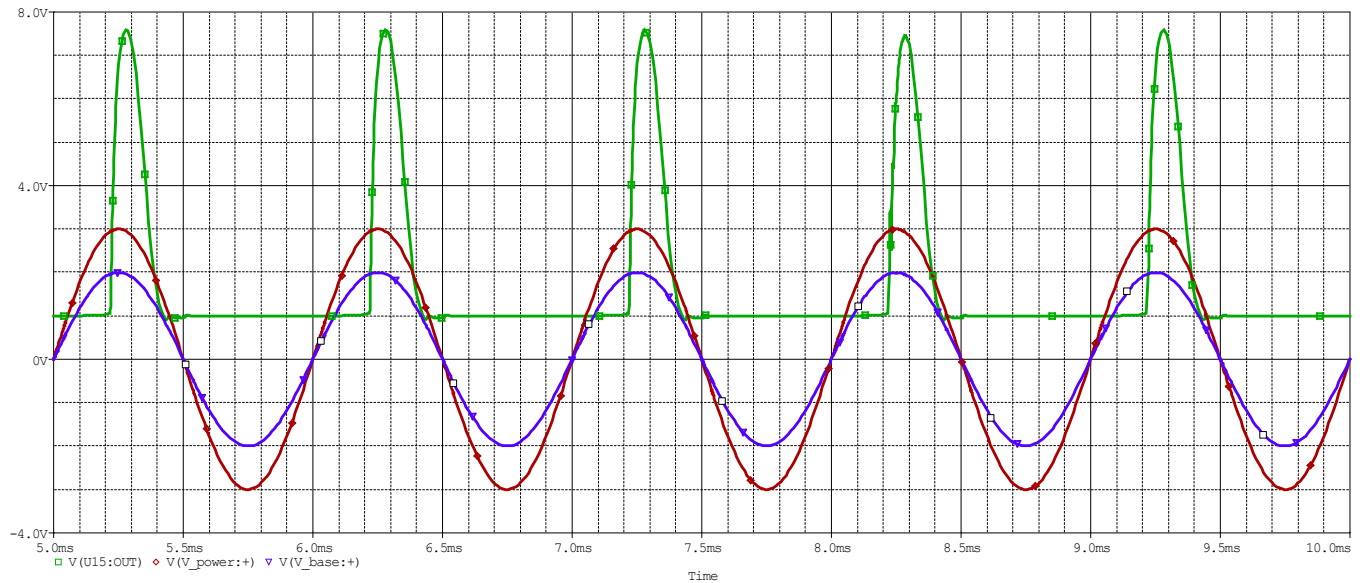
In a similar vein, U11, also configured as a logarithmic amplifier, facilitates the incorporation of an additional input signal, representing the power of the base. Correspondingly, U13, an operational amplifier, undertakes the task of mitigating the impact of extraneous coefficients and factors that tend to emerge within the logarithmic expression.

Our desired output is ultimately derived from the output terminal of U15, the culmination of our carefully orchestrated circuit. Throughout this intricate process, we judiciously employed a combination of inverting amplifiers, logarithmic amplifiers, and exponential amplifiers to effectively counteract and nullify the influence of unwanted factors and coefficients that arise during the calculations.

Output:

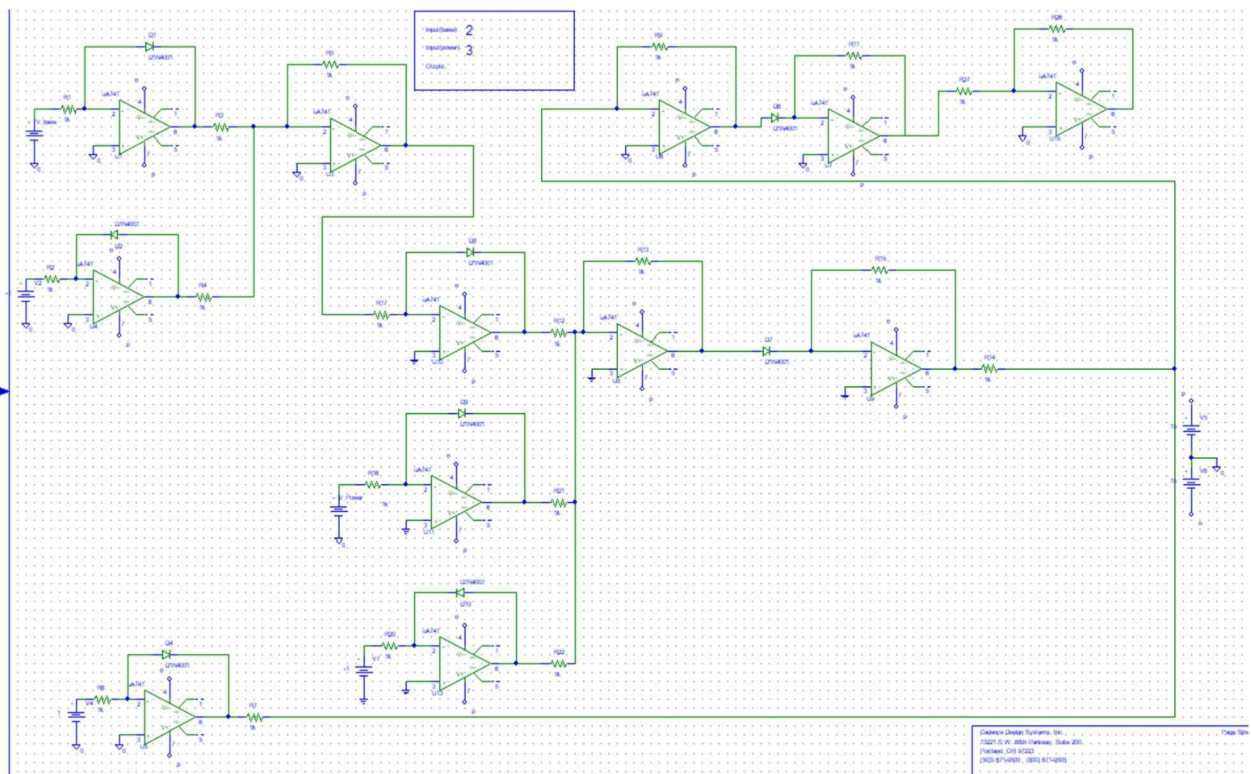
Input base: 2V p Sine wave, 1kHz

Input power: 3V p Sine wave, 1kHz



Here, blue and red are the input & green is output graph.

CIRCUIT & INPUT: (DC)



Output:

Input(base): 2
Input(power): 3
Output: 8.003V

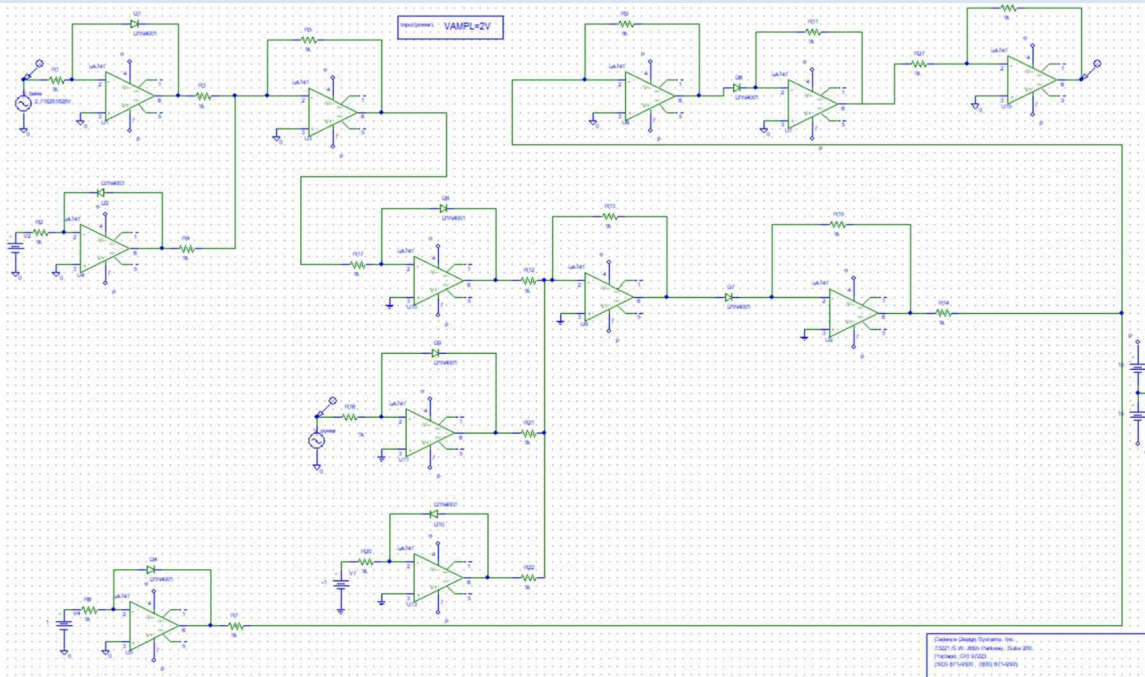
Input(base): 4
Input(power): 1.5
Output: 7.994V

LIMITATION

If input base ≥ 15 , input power ≥ 15 or $v_b^{V_p} \geq 15 V$ then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. This circuit is not suitable for 2-digit input. At times, our circuit may yield decimal values that do not perfectly align with the results expected from a real-life calculator. This particular issue is of paramount importance and warrants rigorous attention and optimization to ensure that our circuit consistently produces precise outcomes that match the expectations of real-life calculations.

EXPONENTIAL:

CIRCUIT & INPUT: (AC)

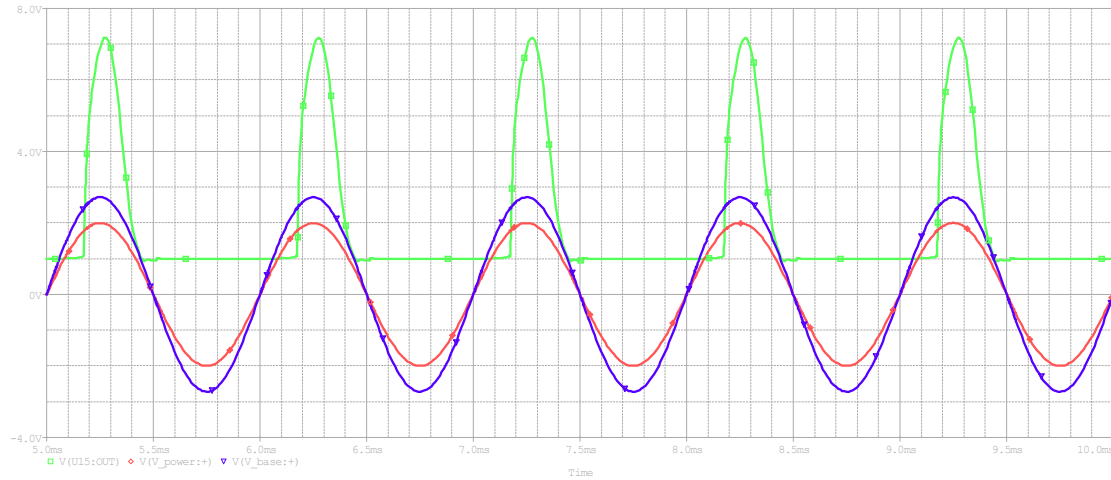


EXPLANATION

Here bias voltage is 15 V and -15 V. In the context of our power calculation circuit, a notable feature emerges when we establish a fixed input base value, specifically $e = 2.718281828$. This deliberate choice empowers our circuit to effectively compute exponential values with precision and consistency.

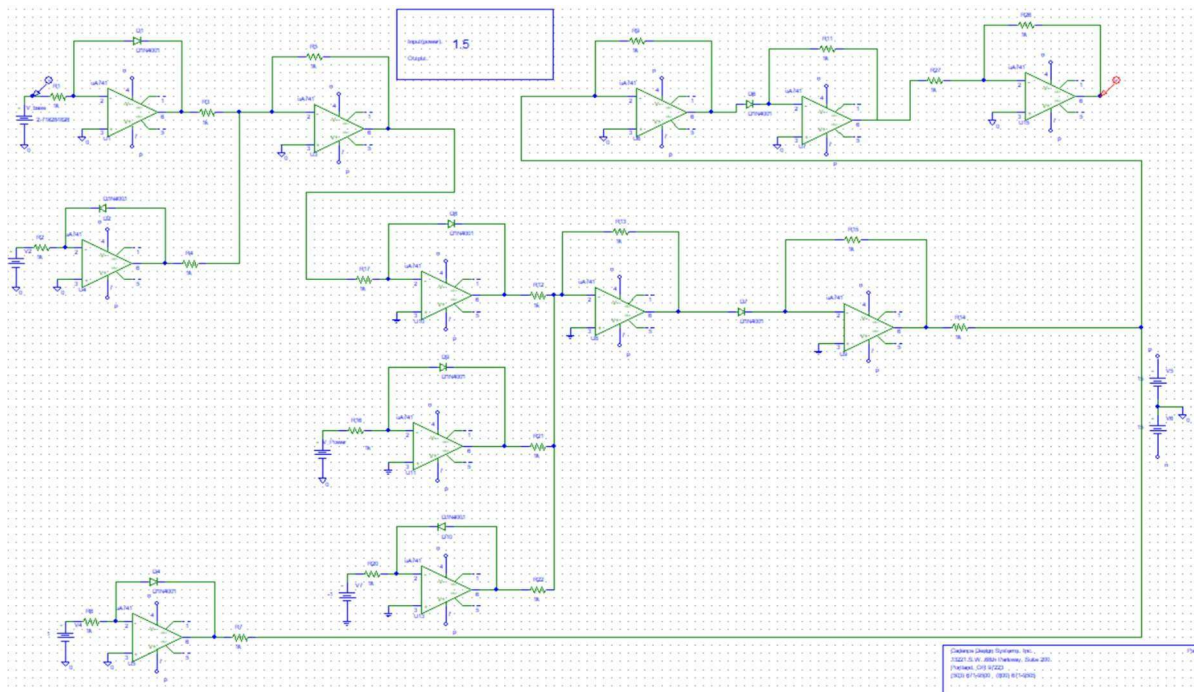
Output:

Input power: 2V p Sine wave, 1kHz

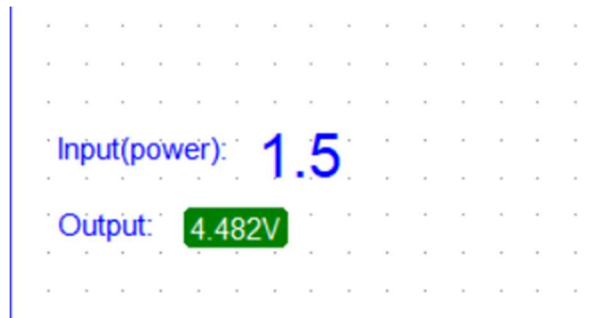


Here, blue and red are the input & green is output graph.

CIRCUIT & INPUT: (DC)



Output:

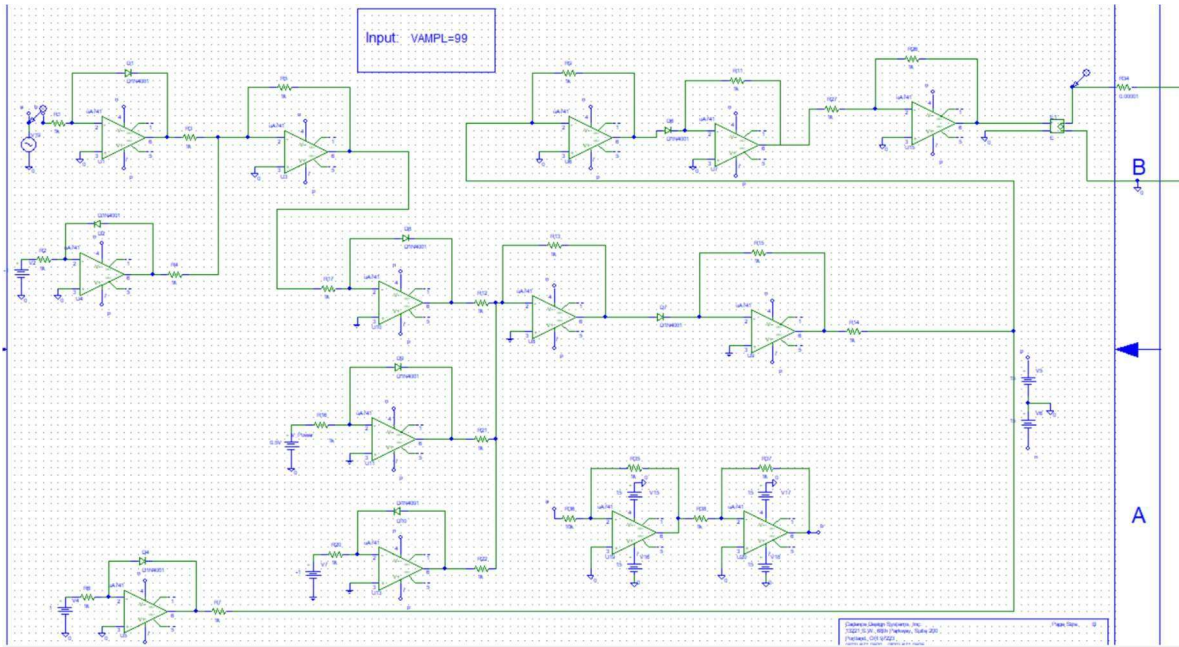


LIMITATION

If input power ≥ 15 or $e^{vp} \geq 15 V$ then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. This circuit is not suitable for 2-digit input. The input power can be up to 2.7V. At times, our circuit may yield decimal values that do not perfectly align with the results expected from a real-life calculator. This particular issue is of paramount importance and warrants rigorous attention and optimization to ensure that our circuit consistently produces precise outcomes that match the expectations of real-life calculations.

ROOT:

CIRCUIT & INPUT: (AC)



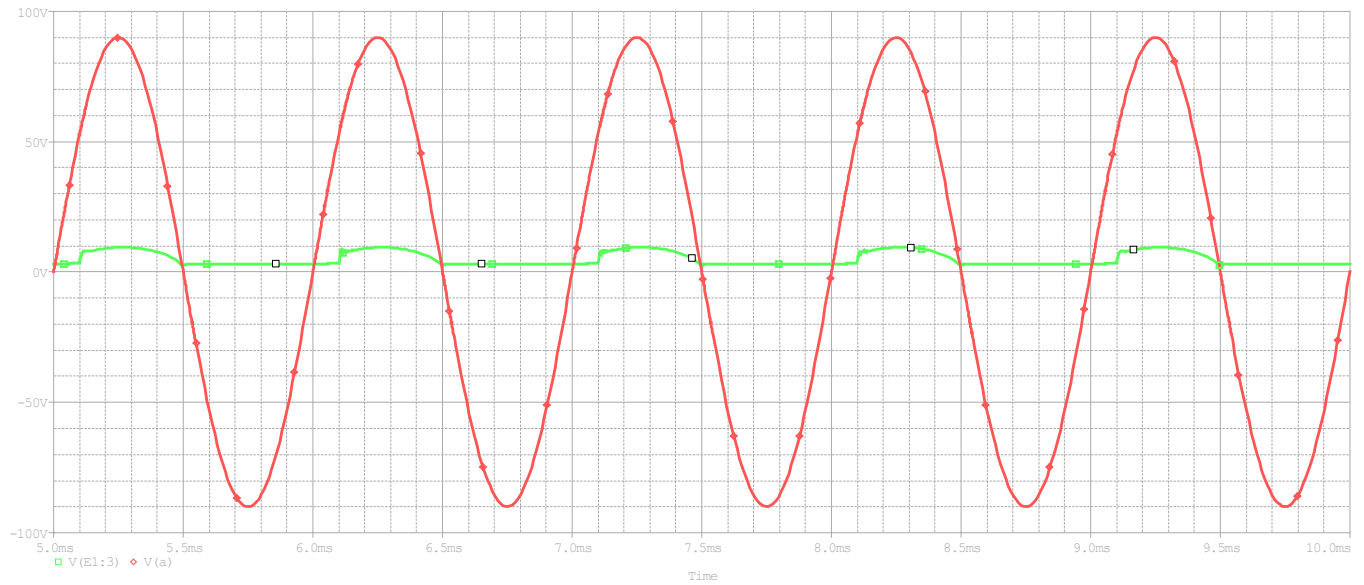
EXPLANATION

Here bias voltage is 15 V and -15 V. In the context of our power calculation circuit, a notable feature emerges when we establish a fixed input power value, specifically at 0.5V. U19 serves as inverting amplifiers, effectively dividing the input signal by 100 and U20 serves as an inverting amplifier making the signal positive again. This design choice is pivotal in allowing our circuit to accommodate 2-digit input values while preventing the operational amplifiers from entering saturation.

Subsequently, the output signal undergoes a substantial 10-fold amplification through a voltage-controlled voltage source to deliver the desired result.

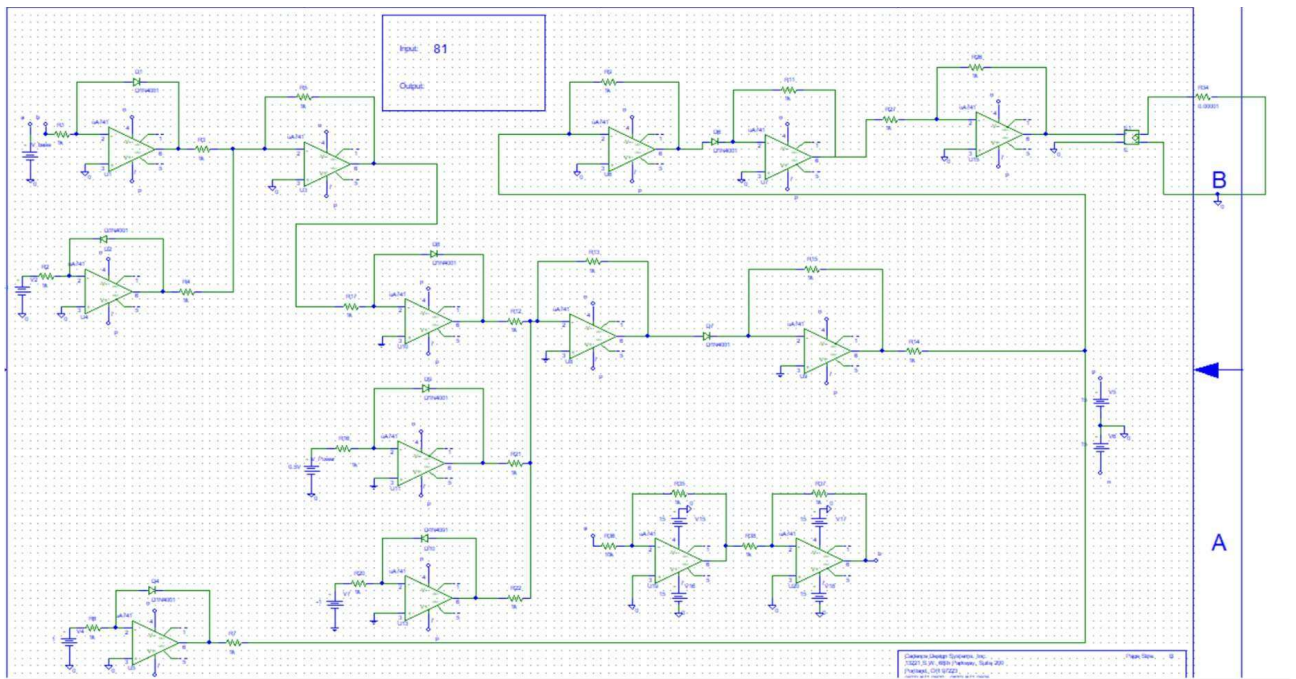
Output:

Input base: 90V p Sine wave, 1kHz



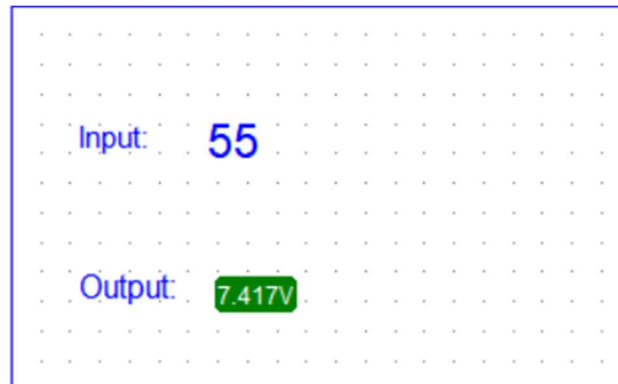
Here, red is the input & green is output graph.

CIRCUIT & INPUT: (DC)

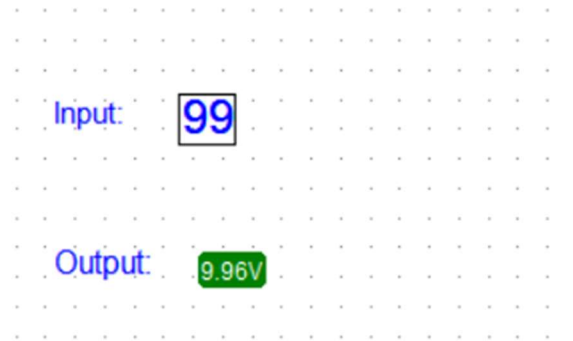


Output:

Case 1:



Case 2:



LIMITATION

If $\frac{\text{input power}}{100} \geq 15$ or $\sqrt{\frac{\text{input power}}{100}} \geq 15$ V then the operational amplifier will reach saturation and the output will be incorrect. This is the biggest limitation of our design. At times, our circuit may yield decimal values that do not perfectly align with the results expected from a real-life calculator. This particular issue is of paramount importance and warrants rigorous attention and optimization to ensure that our circuit consistently produces precise outcomes that match the expectations of real-life calculations.

FUTURE WORK

While our project has successfully achieved its primary objectives, it is important to acknowledge the limitations imposed by both software restrictions and specific operational amplifier (op-amp) configurations. However, this project offers exciting prospects for further development and expansion. Our project can serve as a foundational platform for the development of a scientific calculator. By incorporating advanced mathematical functions, trigonometric calculations, and statistical features, we can create a versatile tool for scientific and engineering applications. To enhance the practicality of our calculator, future efforts can focus on expanding its input capabilities. This involves accommodating input values with at least 12 digits, ensuring precision in handling decimal values. This improvement would make our calculator more suitable for complex calculations in various domains. Our project has the potential for commercial viability. With its efficient design that minimizes the need for extensive equipment and maintains safety standards, there is an opportunity to bring this calculator to market. Addressing the input precision and expanding its functionality can make it an appealing product for a wide range of users.

In conclusion, our project lays a strong foundation for the development of a versatile calculator with significant future prospects. By addressing the identified limitations and expanding its capabilities, we can create a commercially profitable and highly functional tool that caters to diverse mathematical and scientific needs.

DISCUSSION:

Throughout the course of this project, our central objective revolved around the creation of an analog calculator, specifically tailored to accept 2-digit inputs. Leveraging operational amplifiers (op-amps) while maintaining a stringent focus on preserving accuracy posed as our guiding principle. One of the most significant challenges encountered was the task of circuit modification to facilitate 2-digit inputs without surpassing the bias voltage limits. We achieved this successfully across all operations, with the notable exceptions being exponential and power calculations.

In the realm of practical implementation using PSPICE, our op-amps were indeed equipped with offset currents, which introduced an inherent impact on result accuracy. To mitigate this effect, we dedicated considerable efforts to minimizing its influence, thereby elevating the precision of our outcomes.

Our project encompassed both AC differentiator and AC integrator circuits, where we discovered that only input signals oscillating at a 1kHz frequency could guarantee correct outputs. Deviations from this frequency necessitated adjustments to the capacitor (C) value, ensuring that the desired gain of 1 was maintained.

Importantly, our project excelled in achieving a remarkably high level of accuracy. It not only met but surpassed our initial goal of accommodating 2-digit inputs, extending its capability to handle a range of up to 3 digits. This enhanced flexibility opens up diverse applications and scenarios where more extensive calculations are required.

In conclusion, our project represents a successful endeavor in the development of an analog calculator with a strong focus on precision and flexibility. While encountering challenges related to offset currents and frequency dependencies, we diligently addressed these issues and expanded the operational boundaries beyond our initial targets. The outcome is a versatile tool that holds great potential for various computational tasks, demonstrating the power of analog computation in a digital age.